Research Roadmap

Quantum-well solar cells, thermophotovoltaics and nonlinear intersubband devices

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Status. Paper 1 (N1, sign-reversal quasi-phase matching) is written; its pre-submission checks are parked in LATER_STAGE.md and in the Research atlas section "Parked for a later stage". The Research atlas section ranks 51 paper opportunities (45 need no experiment) and recommends a six-paper programme: qwkit software paper, saturation bounds, many-body second-order response, robust inverse design, switchable ISB thermal emitters, and $LaInO_3/BaSnO_3$ oxide quantum wells.

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Research Roadmap: Quantum-Well Solar Cells, Thermophotovoltaics & Nonlinear Intersubband Devices

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Summary

The best openings for a quantum-well group are not efficiency records but the five openings below: three questions the chapter itself leaves open and two cross-thread ideas we found no attempt at. Since the book went to press, records have moved a long way (perovskite/Si tandem 35.5%, strain-balanced QW 3J 39.5% at 1 sun, TPV 44% at 1435 °C, DFG-QCL ~2 mW pulsed THz), while the physics questions underneath have not.

1. **Do quasi-Fermi levels stay flat across the wells?** Measured once at ~5–18 meV suppression in 2010; never re-tested on today's >40% radiative-efficiency QW cells (Thread B).
2. **Narrow-gap TPV for 600–1100 °C heat.** High-temperature TPV is solved; this window is not. An air-bridge interband cascade cell joins two proven ideas (Thread D).
3. **True sign-reversal quasi-phase matching of intersubband $\chi(2)$.** Only Stark-modulated QPM has been shown (2006); a bias-polarity scheme in symmetric coupled wells could flip the sign electrically (Thread E, N1).
4. **A ratchet intermediate-band cell designed with the ADQW nonlinear toolkit**, the one route theory says can beat Shockley–Queisser at any sub-gap absorptivity (Thread C, N2).
5. **Interband cascade thermoradiative and bidirectional cells**, using multistage voltage to fight the non-radiative losses that cap today's 'night-time solar' diodes (N3).

Read the Framing table first, then go straight to any thread; each ends with its contribution points. The phased roadmap and gate criteria near the end turn this into a 5-year plan.

Framing: every thread attacks one named loss

The chapter's four fundamental losses (Sec. 15.1) are the cleanest way to sort the field: each device concept exists to remove one of them, and each open problem is a reason it has not yet done so. The same accounting applies to TPV, with the emitter temperature replacing 5800 K. Nonlinear intersubband devices sit on a parallel axis where the trade is $\chi(2)$ against absorption (Eqs. 15.39–15.42).

Loss (chapter ref.)	Size at 1 sun, ~1.3 eV gap	Concepts that attack it	Status in 2026
Carnot, Eq. 15.9	Small; $\eta_C \approx 95\%$	None practical; bounds everything	Fixed by physics
Solid-angle mismatch, Eq. 15.10	≈ 0.28 V off Voc	Concentration, angle-restricted emission, photon recycling	Mature (concentration); angle restriction under-explored
Sub-gap transmission	Largest single loss	Multi-junction, QW/IB cells, up-conversion; TPV spectral recuperation	MJ mature; IB stalled; TPV recuperation now dominant lever
Thermalization	Second largest	Multi-junction, hot-carrier, MEG/singlet fission, energy-selective contacts	MJ mature; hot-carrier still pre-device
Non-fundamental (SRH, Auger, resistance, shading)	Sets the gap to records	Better materials, thin absorbers + light trapping	Incremental but where most papers live

The chapter's own unresolved question, whether quasi-Fermi levels stay flat across a QW-loaded junction, is the hinge between Thread B and Thread C. It is listed as an explicit open problem below because it decides whether QW cells are a current-matching tool or a genuine beyond-Shockley–Queisser route.

Thread A: single and multi-junction cells

Multi-junction is the only thread that has actually beaten the single-junction limit at scale, and quantum wells already sit inside two of its records. The chapter's figures (GaAs ≈ 29%, Si 26.7%, 4J 37.8% at 1 sun, 46% concentrated) are now out of date; current reference points are below.

Device	Efficiency	Holder, date	Note
Perovskite/Si tandem, 2T	35.5%	LONGi, ESTI-certified, mid-2026 (PV Tech)	35.2% listed in Efficiency Tables v68 (May 2026)
III-V 4J, concentrated	47.6% at 665 suns	Fraunhofer ISE (SurgePV summary)	Absolute PV record
GaInP/GaAs/GaInAs 3J with SB-QW, 1 sun	39.5%	NREL (NREL)	Up to 300 strain-balanced GaInAs/GaAsP pairs
GaInP/GaAs 2J with 80 QWs, 1 sun	32.9%	NREL, Adv. Energy Mater. 11, 2002874 (NREL)	First record MJ using strain balancing
Perovskite/perovskite/Si 3J	28.7%	KAUST-led (pv magazine)	Far below its limit; open field

Open problems

- Perovskite top cells: long-term stability, halide segregation in 1.6–1.8 eV wide-gap compositions, and the Voc deficit relative to the radiative limit (Eq. 15.14). The 2T perovskite/Si limit is quoted near 43%; records are about 8 points short.
- III-V cost: substrate reuse, HVPE growth, and III-V-on-Si tandems that keep the III-V top cells but drop the Ge/GaAs substrate.
- Spectral and temperature robustness of current-matched stacks (the chapter's diurnal and seasonal mismatch point) — still mostly handled by over-design, not by adaptive devices.
- Triple-junction perovskite/perovskite/Si and III-V/Si: few groups, large gap to theory.

Where a QW group fits. Thread A is not where a small group wins on absolute efficiency. The opening is supplying engineered absorbers to it: strain-balanced QW superlattices as the ~1.0–1.15 eV bottom junction (Fig. 15.7) and as a current-matching knob in III-V top cells on Si.

Thread B: quantum-well solar cells

Strain-balanced QWs have won as a bandgap-tuning tool (32.9% 2J, 39.5% 3J at 1 sun) but have not been shown to beat the bulk-equivalent Voc in a way that matters. The chapter's question — do quasi-Fermi levels stay flat across the wells? — has a partial experimental answer: absolute EL/PL measurements found a quasi-Fermi-level splitting suppressed in the wells by about 18 meV (1 well) and 5 meV (5 wells) relative to the bulk, possibly offset by hot carriers in the wells (Mazzer et al., 2010). That is small, and it has not been revisited on modern high-radiative-efficiency devices.

State of the art

- NREL grew up to 300 strain-balanced GaInAs/GaAsP pairs to reach a ~1.33 eV-class absorber, with good collection and low nonradiative loss; their 80-QW single junction kept Voc above 1.04 V via a rear heterojunction (NREL, Adv. Energy Mater.).
- The same programme reported external radiative efficiency above 40% in a QW single junction (NREL news) — high enough that photon recycling and angular emission now matter.
- 2025 work on InGaAs/InAlAs hot-carrier QWs found Voc tracking confinement, but photocurrent coming mostly from the barriers rather than the wells (Aydin et al., arXiv 2507.04112).

Open problems

1. **The QFLS question, revisited at high ERE.** Is the well QFLS suppression still ~5 meV in a >40% ERE, 80–300 well device, and does it depend on well depth, escape time and field? Hyperspectral absolute PL plus EL reciprocity on graded test structures would settle it.
2. **Well count versus absorption.** Each pass through a well absorbs only 0.6–0.7% (Sec. 15.2), forcing 80–300 wells and a deep strain budget. Resonant-cavity or textured-rear designs that cut the well count 3–5× would reduce growth time, interface recombination and field-collapse risk.

3. **Collection in thick i-regions.** Hundreds of wells need very low background doping (Eq. 14.17); the trade between field-assisted escape and recombination is still tuned empirically.
4. **QWs in the wide-gap subcell.** InGaAsP wells in In0.49Ga0.51P (chapter, p. 597) are far less developed than the GaAs-side wells; this is a clean current-matching lever for 2J/3J and for III-V-on-Si tops.
5. **A ~1.0 eV QW 'material' bottom junction** on GaAs (Fig. 15.7) competing with metamorphic and dilute-nitride alternatives — strain and absorption limits at that depth are not well mapped.

Contribution points. Items 1 and 2 are tractable for a group with MBE/MOVPE access and absolute PL; item 1 alone would be a widely cited result because it underpins both QW and hot-carrier claims.

Thread C: intermediate-band, ratchet and hot-carrier cells

Both concepts remain at proof-of-principle, and theory now says *why* the simple versions fail: a plain intermediate band cannot satisfy voltage preservation and current matching at once, and a ratchet removes that conflict (Delamarre et al., Prog. Photovolt. 2018). Quantum wells are argued to suit ratchets better than the quantum dots most groups have used (Delamarre et al., IEEE JPV 2018). This is the most QW-native opening in the whole chapter.

Intermediate band and ratchet (Fig. 15.10)

- Without a ratchet, an IB cell cannot beat Shockley–Queisser at 1 sun unless both sub-gap absorptivities exceed about 36%; with a ratchet it can at any IB absorptivity (Pusch et al., Prog. Photovolt. 2016).
- A UK–Australian quantum-ratchet IB cell showed two-photon photocurrent at 300 K, using an AlAs barrier to stop thermal escape from the ratchet band (pv magazine, 2023). Its gaps were not solar-optimised.
- Okada's group (Tokyo) is pursuing a ratchet IB film using long-lived Er-doped GaAs states plus epitaxial lift-off and light trapping (JST project summary).

Open problems: sub-gap absorptivity (intersubband IB→CB absorption is TM-only and weak at normal incidence — the same $\sin^2\theta$ problem as Eq. 15.26); ratchet depth versus kT backfilling; extraction from the ratchet band; and no device yet that beats its own reference cell under 1 sun.

Hot-carrier cells (Fig. 15.11)

- Limits: >67% under AM1.5 and >86% at full concentration in the radiative limit (Conibeer et al.).
- 2025 Monte Carlo work reproduced slow cooling in type-II InAs/AlAsSb MQWs and confirmed that nanostructuring strengthens the hot-phonon bottleneck (Filipovic et al., Photonics North 2025).
- InGaAs/InAlAs QW devices show hot-carrier QFLS signatures, but photocurrent mostly from barriers (Aydin et al. 2025).
- Thermodynamic analysis shows hot-carrier Voc can exceed the gap through a thermoelectric term (Limpert et al., New J. Phys. 17, 095004).

Open problems: a working energy-selective contact (resonant-tunnelling QW contacts exist but lack transmission and selectivity together); hot-carrier evidence at 1 sun rather than at high intensity or low lattice temperature; and separating hot-carrier Voc gain from simple heating.

Contribution points. (1) A QW ratchet cell built on the chapter's ADQW design toolkit, using strain to set the ratchet offset and a grating or 45° facet to fix TM coupling. (2) A QW resonant-tunnelling energy-selective contact, characterised by its own transmission spectrum, decoupled from the absorber question.

Thread D: thermophotovoltaics

The chapter's 'elusive' goal of ≥20% at ≥1000 °C has been passed twice over: 41–44% is now demonstrated above ~1400 °C, and a company runs a dedicated TPV cell line. The frontier has moved down in temperature and up in power density — exactly where narrow-gap QW and cascade absorbers are needed and where results are still weak.

Result	Efficiency	Emitter	Key idea	Source
InGaAs(P) air-bridge, 1J	44%	1435 °C	97–99% out-of-band reflectance via air gap + Au	Roy-Layinde et al., Joule 8, 2135 (2024)
1.4/1.2 eV III-V tandem	41.1% at 2.39 W/cm²	2400 °C	Band-edge filtering with back reflector	LaPotin et al., Nature (2022), arXiv 2108.09613
1.2/1.0 eV III-V tandem	39.3% at 1.8 W/cm²	2127 °C	Same	LaPotin et al.

Result	Efficiency	Emitter	Key idea	Source
Antora production cells	>40% (claimed)	up to ~1500 °C	2 MW cell line, carbon-block thermal battery	Antora
Near-field InGaAs thin film	6.8% at ~5 kW/m²	1270 K, <100 nm gap	Evanescent coupling	Mittapally et al., Nat. Commun. (2021)
~0.4 eV T2SL interband cascade, 3 stages	9.6%, Voc 0.8 V	Low-T source, cell at 300 K	Multistage thin absorbers	Lotfi et al., Sandia/OU (2017)

Other markers: a 150 nm pillar-defined nanogap device produced 25× the far-field output over a large area (UW–Madison/NREL, Adv. Mater. 2024); a zero-vacuum-gap far-field design reached near-field-class power density at 700–1100 °C (Energy Environ. Sci. 2025); stacked air-bridge tandems are in progress (NSF PAR). The chapter's 7-stage 0.25 eV interband cascade cell (Voc 0.65 V) remains the representative narrow-gap result.

Open problems

1. **The 600–1100 °C gap.** Industrial waste heat and cheaper storage media sit here. Optimal gaps are 0.3–0.6 eV (Fig. 15.13), where SRH/Auger lifetimes are short and Meff ≫ 1 (Eq. 15.16). No cell in this window is anywhere near its radiative limit.
2. **Recuperation for narrow-gap cells.** Air bridges work because out-of-band photons are recycled to the emitter. For 0.2–0.5 eV cells, free-carrier absorption in contacts and substrates eats the out-of-band power; nobody has shown >95% out-of-band reflectance on an antimonide cascade cell.
3. **Series resistance and current density** in near-field operation, where photocurrent densities rise 10–25×.
4. **Thermal isolation** of nanogap devices, and emitter/cell lifetime above 1000 °C.
5. **Cascade-stage current matching** under a blackbody spectrum that shifts with storage state of charge.

Contribution points. A narrow-gap QW or T2SL cascade cell on a lifted-off membrane with an air bridge — pairing Thread B growth skills with the 2024 recuperation result — is the clearest single experiment in this roadmap. It attacks open problems 1, 2 and 5 together.

Thread E: intersubband nonlinear optics

The field has split in two since the chapter's analysis. Waveguide THz difference-frequency QCLs have climbed from a few μW to milliwatt-class pulsed output, and a second branch — intersubband polaritonic metasurfaces — has turned the ADQW $\chi(2)$ into an electrically programmable, sub-wavelength-thin nonlinear element. Both are still limited by the chapter's Eq. 15.37 saturation and by free-carrier loss.

THz DFG-QCLs (Sec. 15.5, Fig. 15.17)

- Room-temperature output up to 1.9 mW pulsed and 0.014 mW cw (Razeghi group review, NSF PAR).
- Monolithic tuning across 2.6–4.2 THz at up to 0.1 mW, and a room-temperature THz harmonic comb at 2.2–3.3 THz (NSF PAR).
- Over 112 μW peak at 1.9 THz in double-metal waveguides with surface-grating outcouplers (NSF PAR).

Intersubband polaritonic metasurfaces

- All-dielectric version: 3 mW/W² and 0.045% SHG efficiency at 6.7 kW/cm² (Sarma et al.); plasmonic versions have exceeded 0.2% (as summarised in Adv. Sci., PMC12904043).
- Electrical control of both SHG magnitude and phase (Sci. Adv., PMC12292913), and Stark-tuned broadband SHG across 8.9–10.6 μm (Yu et al., Nanophotonics 13, 1131 (2024)).
- Two-level intersubband designs reduce intensity saturation relative to three-level ADQWs (Krakofsky et al., Proc. SPIE 12431).
- Mid-IR DFG in a metasurface at 0.13% with $\chi(2)_{eff} \approx 340$ nm/V (UT Austin dissertation).

Open problems

1. **Saturation ceiling.** Eq. 15.37 caps pump intensity at ~10 MW/cm² for typical parameters; metasurfaces hit their best efficiency at kW/cm² and then roll off. Two-level and detuned designs (Eq. 15.40) help but have no agreed figure of merit across waveguide and metasurface geometries.
2. **True sign-reversal quasi-phase matching in intersubband waveguides.** The chapter says periodic poling has never been attempted. A weaker form exists: periodic bias that Stark-shifts resonances in and out of tune along a QCL ridge (Belkin et al., APL 88, 201108 (2006)). That modulates $|\chi(2)|$ rather than flipping its sign, so on/off modulation

recovers at most a quarter of the efficiency of ideal sign-reversal QPM (Fourier amplitude $1/\pi$ versus $2/\pi$). The sign of $\chi(2)$ flips with well order (p. 606), and metasurfaces now show electrical control of $\chi(2)$ phase.

3. **THz extraction.** DFG-QCL cw power is still ~10^{-5} of the mid-IR pump power; free-carrier absorption and Cherenkov out-coupling losses dominate (p. 614).
4. **cw thermal budget** for metasurfaces and DFG cores at useful intensities.
5. **Quantum light.** Intersubband $\chi(2)$ for mid-IR photon-pair generation is barely explored.

Contribution points. Open problem 2 is the most open door in the chapter: a lateral grating that genuinely reverses the sign of $\chi(2)$ on an ADQW waveguide (regrowth of mirrored well order, or a symmetric coupled-well design whose $\chi(2)$ sign follows bias polarity). It extends the 2006 Stark-modulation result with established growth and lithography, and a first sign-reversal demonstration is publishable regardless of absolute efficiency.

Cross-cutting enablers

Three shared capabilities decide whether any thread above can be attacked; building them once serves all five.

Enabler	What it unlocks	Threads
Strain-balanced growth of 100–300 periods (GaInAs/GaAsP, InGaAs/InAlAs) and the 6.1 Å InAs/GaSb/AlSb family	QW absorbers, ratchet cells, ADQWs, cascade TPV	B, C, D, E
Epitaxial lift-off / membrane transfer onto IR-transparent or air-bridge carriers	Photon recycling, out-of-band recuperation, thin cascade cells, near-field gaps	B, D
Absolute, calibrated hyperspectral PL and angle-resolved EL	Direct QFLS maps; tests of flat quasi-Fermi levels; ERE and angular emission	B, C
Calorimetric TPV efficiency measurement (as used by LaPotin et al.)	Honest heat-to-power numbers; near-field and far-field on one footing	D

Enabler	What it unlocks	Threads
Coupled models: detailed balance + drift-diffusion + escape rates for wells; Eq. 15.29 + subband rate equations for χ(2) devices; Monte Carlo for hot carriers	Design before growth; separation of real effects from heating	All

One reporting gap is worth fixing as a community contribution: TPV 'efficiency' is defined differently across papers (cell-only versus radiative heat absorbed versus system). Near-field results in particular use the ratio of electrical output to radiative transfer, which is not comparable with calorimetric far-field records.

Novel approaches worth testing

These are hypotheses, not findings. Each crosses two threads of the chapter, names the loss it attacks, and comes with a cheap first test and a kill criterion. Searches turned up adjacent work for most of them (noted), but not these specific combinations.

N1. Bias-polarity sign-reversal QPM in symmetric coupled wells (Thread E). A symmetric coupled double well has χ(2) = 0 at zero field; an applied field breaks the symmetry, and to first order χ(2) is odd in the field. Interdigitated electrodes of alternating polarity along a passive waveguide would then give true sign-reversal QPM with a lithographically set period, and the Stark shift gives a second knob to fine-tune Δk. *Adjacent work:* Stark-modulated (not sign-reversed) QPM in a QCL, 2006. *First test:* compute χ(2)(F) for symmetric InGaAs/InAlAs pairs with the chapter's Eq. 15.27; check that |χ(2)| at ±F is ≥30% of an optimised ADQW. *Kill if* the field needed exceeds breakdown or kills the resonance linewidth.

N2. A ratchet IB cell designed with the SHG figure-of-merit toolkit (Threads C + E). The ADQW of Fig. 15.14 is already a three-level ladder with engineered z12, z13, z23, τ12 and Δ. Re-purpose it: interband VB→well absorption fills the IB, fast tunnelling into the neighbouring well's lower state is the ratchet, and an intersubband transition lifts electrons to the continuum. The chapter's Eqs. 15.36–15.40 then become a design rule for two-step absorption versus saturation. *First test:* model two-step photocurrent per well versus ratchet offset (target 3–5 kT) and grating-coupled TM absorption. *Kill if* modelled sub-gap absorptivity stays below the ~36% one-sun threshold even with a 1-sun-optimised grating and photon recycling.

N3. Interband-cascade thermoradiative and bidirectional heat engines (Thread D). Thermoradiative diodes now produce measurable power when facing a cold sink, but about 10^5 times less than a solar panel (UNSW, pv magazine Australia); concentrated near-field versions are being modelled (Feng et al., ACS Nano 19, 17357 (2025)). Their weak point is non-radiative recombination at low voltage — exactly what multistage cascade designs mitigate in TPV (Eq. 15.16). One cascade cell could run as TPV facing a hot store and as a thermoradiative generator facing the sky or a cold sink. *First test:* measure negative-illumination power of existing interband cascade TPV or detector structures facing an LN2-cooled plate, as in Matsuura & Handa, arXiv 2402.08253. *Kill if* power density per stage is no better than single-junction InAs or HgCdTe diodes.

N4. Narrowband near-field TPV: phonon-polariton emitters matched to cascade absorbers (Thread D). The chapter's monochromatic-pump example reaches ~85% when the photon energy sits just above the gap. Polar emitters (SiC, hBN) concentrate near-field transfer into a narrow band near 0.1 eV. Pairing them with a 0.1–0.15 eV T2SL cascade cell could give high-efficiency conversion of 300–600 °C heat, the range no current TPV serves. *First test:* fluctuational-electrodynamics calculation of spectral transfer into a measured cascade-cell absorption spectrum versus gap. *Kill if* free-carrier and phonon absorption in the cell's contacts take more than half the transferred power.

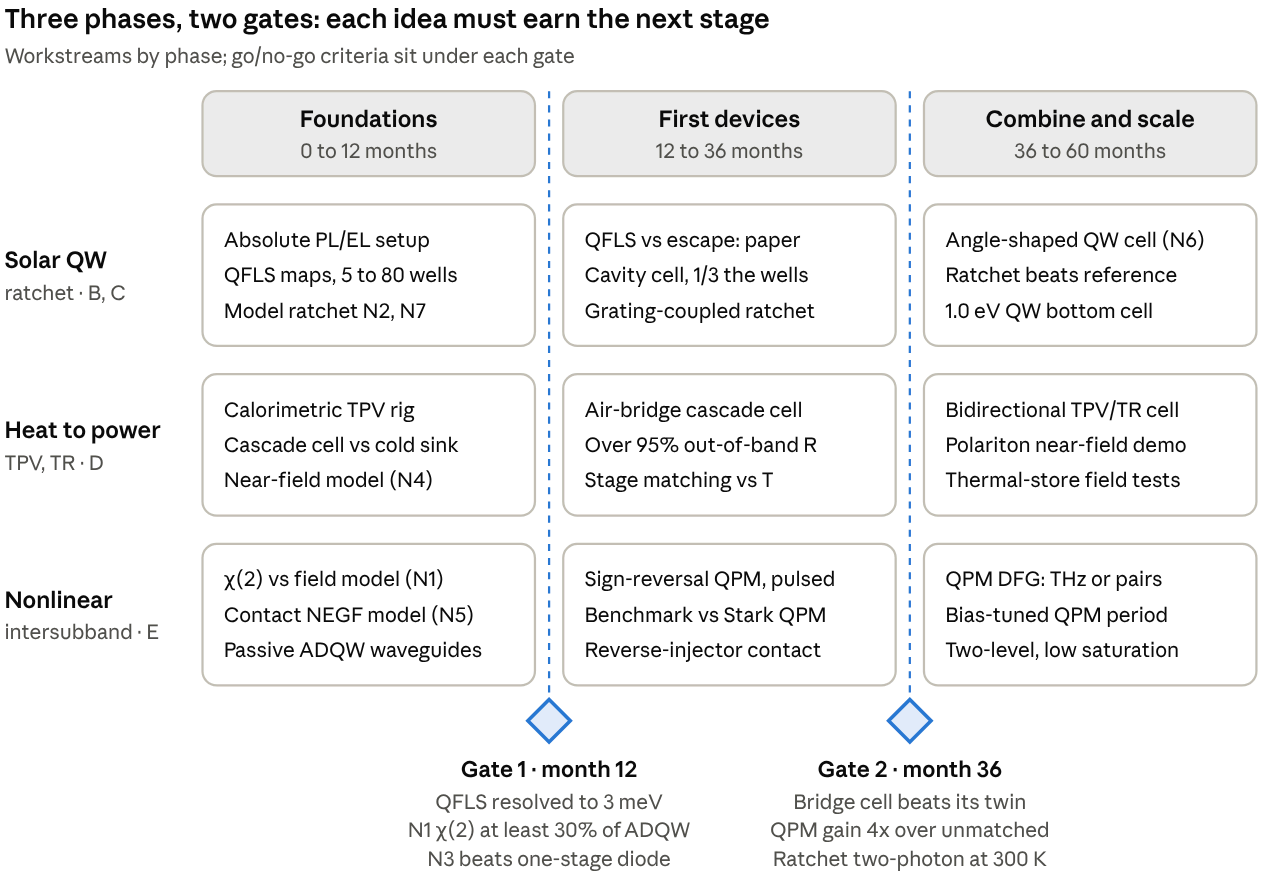
N5. Reverse-QCL injectors as energy-selective contacts (Thread C). QCL designers have decades of practice building LO-phonon-resonant, miniband-filtered injection and extraction at a chosen energy. Running that machinery in reverse is a route to the missing energy-selective contact, and InAs/AlAsSb wells already show a strengthened hot-phonon bottleneck. *First test:* NEGF transmission spectrum of a 3–4-well injector designed for 150–250 meV above the band edge; target width < kT and peak transmission > 0.5. *Kill if* no design meets both.

N6. Strain-shaped angular emission to recover part of the solid-angle loss (Threads A + B). By reciprocity, emission at each angle equals absorptivity at that angle. Heavy-hole transitions couple only to in-plane fields; tensile strain brings light-hole states with a z-polarised component to the band edge. Combining polarisation selection rules with an angular Bragg filter in a >40% ERE QW cell could narrow emission toward the sun's cone, attacking the ≈0.28 V of Eq. 15.10. Angle restriction has been proposed for bulk cells; doing it with QW selection rules has not, to our search. *First test:* angle-resolved EL on existing compressive versus tensile QW cells. *Kill if* the edge emission pattern differs by <10% between strain states.

N7. Self-current-matching QW subcells (Thread B). In a two-terminal stack, the current-limiting subcell is driven toward short circuit. If its QWs sit in the junction field, that shift increases the field, red-shifts the QW edge via the Stark effect and speeds escape — raising its photocurrent. A designed negative-feedback loop could soften the spectral-mismatch penalty the chapter attributes to diurnal and seasonal variation. *First test:* drift-diffusion plus Stark-shifted absorption model of a GaInP/GaAs-QW stack across AM1.5 and red-rich spectra. *Kill if* the annual energy-yield gain is <0.5% relative.

Phased roadmap with go/no-go gates

Year one is mostly measurement and modelling, so most ideas are tested before growth time is committed; devices follow only for ideas that pass Gate 1.



phased roadmap · 3 workstreams, 3 phases, 2 gates

At each gate, a workstream that misses its criterion drops to modelling-only and its growth time moves to the strongest remaining item. Criteria are deliberately binary so a gate review takes an afternoon.

For a team without a lab, phases 1 and 2 run as calculations: each 'device' box becomes a design-and-simulation deliverable, and fabrication moves to phase 3 only for ideas that pass both gates on paper.

Risks, and where a small group can win

A small group wins on questions large labs have left unanswered, not on headline efficiency. Thread A records are set by NREL, Fraunhofer ISE and module makers with production lines; competing there is a poor use of a QW group's time.

Risk	Where it bites	Mitigation
Growth access: 100–300-period strain-balanced stacks or antimonide cascades take weeks of MBE/MOVPE time	B, C, D	Start with modelling-led items (N1, N2, N5, N7); partner with an NREL-, Sandia- or university-foundry grower for wafers
Hot-carrier and QFLS signals confused with heating	B, C	Report lattice temperature independently; use the thermoelectric-voltage separation from Limpert et al.
TPV efficiency numbers not comparable	D	Calorimetric measurement from the start; state the efficiency definition in every result
Intersubband saturation and heating cap $\chi(2)$ devices below useful cw power	E	Target pulsed or low-duty demonstrations and proof of QPM physics, not absolute power
Fast-moving records make the landscape stale	A, D	Re-check records against the Efficiency Tables and TPV literature every 6 months

Best openings for a small, QW-focused group, re-ranked for a team working without growth or a lab (decided 10 Oct 2026): (1) sign-reversal intersubband QPM, N1 — fully computable, now under way; (2) ratchet design with the ADQW toolkit, N2; (3) narrowband near-field TPV transfer, N4; (4) reverse-QCL energy-selective contact transmission, N5; (5) self-current-matching energy-yield model, N7; (6) a predictive model

of QFLS suppression versus escape time for Thread B, testable against published PL data. Items needing fabrication or measurement — the air-bridge cascade cell, N3, N6 and all of Thread A — are parked until a result justifies approaching a growth collaborator.

First study: N1 study: sign-reversal QPM

How results become papers: Publication framework

References

Base text and its suggested reading

- Vurgaftman, Lumb and Meyer, *Bands and Photons in III–V Semiconductor Quantum Structures*, Ch. 15 (Oxford University Press, 2021).
- J. A. Nelson, *The Physics of Solar Cells* (Imperial College Press, 2003).
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- A. Yariv, *Quantum Electronics*, 3rd edn. (1988); R. Boyd, *Nonlinear Optics* (1992).
- J. R. Meyer et al., IEEE J. Quantum Electron. 31, 706 (1995); K. Fujita et al., Nanophotonics 7, 1795 (2018).

Current-state sources (linked in the text)

- Solar records: PV Tech, LONGi 35.5%; SurgePV records timeline; NREL 39.5% 3J QW; NREL 32.9% 2J QW.
- QW and hot-carrier physics: Mazzer et al. 2010 QFLS; Aydin et al. 2025; Filipovic et al. 2025; Limpert et al. NJP 2015; Conibeer et al..
- Intermediate band and ratchet: Delamarre et al. Prog. Photovolt. 2018; Delamarre et al. IEEE JPV 2018; Pusch et al. 2016; quantum-ratchet cell at 300 K; Okada JST project.
- TPV and thermoradiative: Roy-Layinde et al. Joule 2024; LaPotin et al.; Antora; Mittapally et al. 2021; UW–Madison nanogap TPV; zero-vacuum-gap TPV, EES 2025; Lotfi et al. cascade TPV; UNSW thermoradiative diode; Feng et al. ACS Nano 2025; Matsuura & Handa 2024.
- Nonlinear intersubband: DFG-QCL review; tunable DFG and THz comb; double-metal DFG; Belkin et al. Stark QPM 2006; Sarma et al. all-dielectric metasurface; complex-amplitude SHG control; local-to-nonlocal SHG; Yu et al. Nanophotonics 2024; Krakofsky et al. SPIE 2023; UT Austin metasurface DFG.

Figures were taken from search-indexed abstracts and news pages; confirm against the primary papers before citing them in a proposal.

N1: sign-reversal quasi-phase matching with field-induced intersubband χ(2)

Oct 10, 2026 · computational study, no fabrication

Verdict: N1 passes Gates 1 and 2, with a narrower use case than first proposed. A mirror-symmetric InGaAs/InAlAs triple well (58/14/34/14/58 Å) behaves as an ADQW whose orientation is set by bias polarity, reaching 55% of the best ADQW's χ(2) at 50 kV/cm and 300 K (65% at 77 K). The gate was 30%; 50% ties an ideally Stark-switched ADQW. Propagation modelling shows the advantage — 1.9–3.2× over Stark-modulated QPM — appears only in long, lightly doped waveguides, because at high doping absorption keeps devices shorter than one coherence length. Gate 2 (self-consistent Hartree, depolarisation, error budget) was passed on 10 Oct 2026: 53–61% across model levels, and never below 36% under any tested error.

The question and the gate

N1 claims that a mirror-symmetric quantum-well stack inside a waveguide, biased with electrodes whose polarity alternates every coherence length, gives true sign-reversal quasi-phase matching (QPM) of a resonant intersubband χ(2). The physics is simple: mirror symmetry forces $\chi^{(2)} = 0$ at zero field, and reversing the field is the mirror operation, so $\chi^{(2)}(-F) = -\chi^{(2)}(F)$ exactly.

That symmetry also brings a side benefit no other intersubband QPM scheme has. Subband energies and dipole magnitudes are identical at +F and −F, so the linear absorption and refractive index are the same in every segment. There is no parasitic index grating and no Bragg reflection — only the sign of χ(2) changes.

Gate 1 (from the roadmap): the best symmetric design must reach $|\chi^{(2)}| \geq 30\%$ of an optimised asymmetric double quantum well (ADQW) at the same pump wavelength (10 μm), electron density (10^{18} cm^{-3}) and linewidth (10 meV), at a field that keeps the upper subbands bound. For context, ideal sign reversal has a QPM amplitude factor of $2/\pi$ and ideal on/off modulation $1/\pi$, so a symmetric design at 50% of the ADQW only ties a perfectly Stark-switched ADQW. Real Stark modulation is partial and adds an index grating, which is why the roadmap set the bar lower, at 30%. Results are reported against both lines.

Kill criteria: the required field causes tunnelling escape broader than the 10 meV linewidth, or screening by the electrons cancels most of the applied field.

Prior art: what exists and what is new

Every ingredient of N1 has been shown separately. What we found no record of is the combination: in-plane propagation, resonant intersubband χ(2), and electrically set sign reversal.

Work	What it showed	How N1 differs
Fejer et al., PRL 62, 1041 (1989)	Biased symmetric GaAs wells: χ(2) = 28 nm/V at 36 kV/cm, with sign set by bias direction	Uniform bias, no phase matching
Fiore et al., APL 70, 2655 (1997)	QPM by growing asymmetric wells with reversed orientation every coherence length	Growth-defined, vertical surface-emitting geometry, interband/excitonic
Bouchard et al. (2000); Zeaiter et al., OL 28, 911 (2003)	QPM by intermixing that switches χ(2) off in alternate domains	Modulates magnitude, permanent, near-IR
Belkin et al., APL 88, 201108 (2006)	Periodic bias Stark-detunes intersubband resonances along a QCL; up to 10× gain	Modulates magnitude and creates an index grating
Timurdogan et al., arXiv 1603.04515; Vecchi et al., Proc. SPIE 11364	Field-induced χ(2) from χ(3) in silicon, sign-reversed by alternating p-i-n junctions	Non-resonant χ(3) origin, ~41 pm/V
Ge/SiGe ACQW proposal (ECS 2022)	Waveguide ADQW SHG with QPM by corrugation, ~10% predicted	Geometric grating, not χ(2) sign

So the claim N1 can honestly make is narrower than first written: the silicon EFISH scheme transplanted onto a resonant intersubband χ(2) that is orders of magnitude larger. Calling it 'never attempted' would be wrong.

Model and checks

A conduction-band, 1D effective-mass model for InGaAs/InAlAs lattice-matched to InP, written for this study (code attached in chat).

- **Materials:** In0.53Ga0.47As wells (m* = 0.043, Eg = 0.75 eV), In0.52Al0.48As barriers (m* = 0.075, Eg = 1.45 eV), conduction-band offset 0.52 eV. Standard literature values, not re-fitted.
- **Solver:** BenDaniel–Duke finite differences on a cell-centred grid; Kane nonparabolicity $m(E) = m^*(1 + (E - V)/Eg)$, solved per state by index tracking; states Löwdin-orthonormalised; only states with ⩾80% probability in the active region are kept.
- **$\chi(2)$:** the resonant term of the book's Eq. 15.25 summed over the full ladder (6 states), ground subband only populated, with diagonal terms entering as $(z_{nn} - z_{11})$. Linewidth Γ = 10 meV; N = 10^{18} cm^{-3}; TM coupling ($\sin\theta = 1$).
- **Neglected:** depolarisation and Hartree shifts, non-resonant permutation terms, valence-band mixing, finite temperature. These shift resonances by a few meV and act equally on both structure types, so the *ratio* tested at the gate is more robust than either absolute value.

Check	Result
100 Å well, 50 eV walls, parabolic	E1 = 82.9 meV vs 87.5 meV analytic; the 5% gap equals the expected wall penetration
z12 of that well	18.5 Å vs 18.0 Å analytic
Book Fig. 15.14 ADQW (64/18/33 Å)	E21 = 106 meV, E31 = 225 meV: a near-equal ladder, resonant at ~11 μm in this model vs 10 μm in the book
Symmetric double well, F = 0	$\chi(2) = 0$ exactly
Same well, F = ±50 kV/cm	$\chi(2) = \mp(61.5 + 11.4i)$ nm/V: exactly odd in F

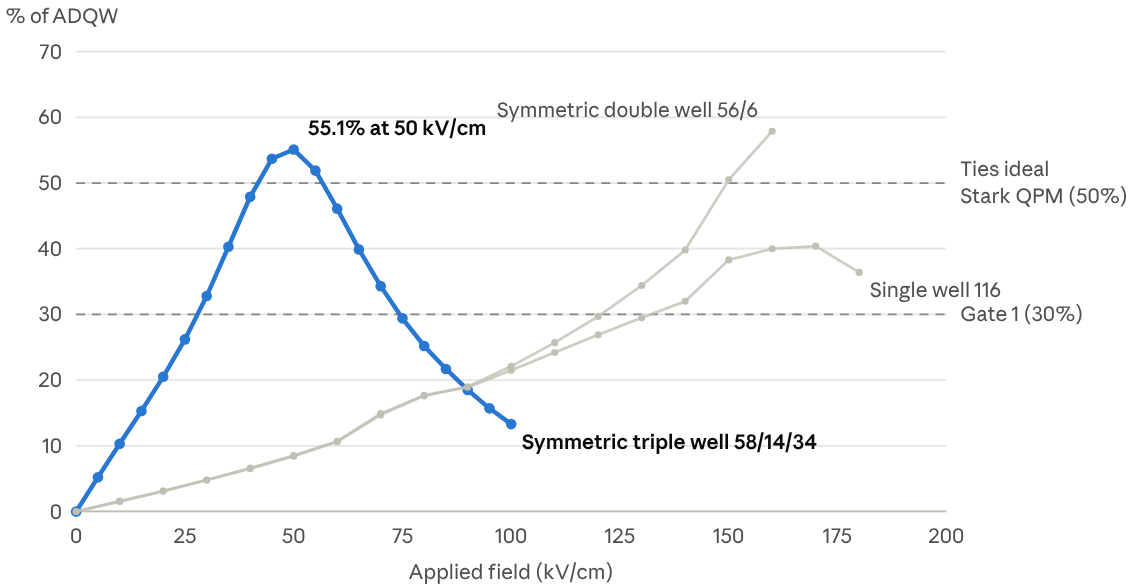
Getting the symmetry checks to pass took three fixes: a cell-centred grid, index-tracked nonparabolic states, and orthonormalisation. Each earlier version produced a spurious $\chi(2)$ at zero field, which is the error a reviewer would look for first.

Results: the gate is passed by one design family

A symmetric triple well (58/14/34/14/58 Å) at 50 kV/cm reaches |χ(2)| = 64 nm/V, 55% of the best ADQW found (64/10/30 Å, 117 nm/V), at 300 K. At 77 K it reaches 79 nm/V, 65%. The simpler symmetric designs need far more field: the double well passes 30% at ~125 kV/cm and 50% at 150 kV/cm, and the single well passes 30% at ~135 kV/cm but never reaches 50% before its third subband unbinds.

A symmetric triple well reaches 55% of the ADQW at 50 kV/cm

|χ(2)| as % of the best asymmetric double well, 10 µm pump, 300 K, same electrons per period



computed with qwchi2.py · InGaAs/InAlAs on InP, Γ = 10 meV, 5×10¹¹ cm⁻² per period, escape broadening < 1 meV

Why the triple well works. At zero field the two outer wells form a near-degenerate pair only 12 meV apart. A modest field pushes the electron into one outer well, and the structure then behaves like an ADQW — outer well, 14 Å barrier, centre well — whose orientation follows the field polarity. In effect it is an electrically reversible ADQW. Single and double wells have no easily polarised ground state, so their χ(2) grows roughly linearly with field until a resonance is reached. Each curve ends where an upper subband stops being bound.

Design point, 50 kV/cm, 300 K	Triple well	Best ADQW (0 V)
Ladder E21 / E31 / E41 (meV)	55 / 126 / 250	122 / 248 / –
Resonant path	1→3 (pump), 1→4 (SH)	1→2 (pump), 1→3 (SH)

Design point, 50 kV/cm, 300 K	Triple well	Best ADQW (0 V)
	$\chi(2)$	(nm/V)
Second-subband occupation	15%	1%
Pump absorption $\alpha(\omega)$, material (cm^{-1})	729	2,162
SH absorption $\alpha(2\omega)$, material (cm^{-1})	1,275	978
Electron density (cm^{-3}), 5×10^{11} cm^{-2} per period	1.3×10^{17}	1.6×10^{17}

Growth tolerance. One monolayer (±3 Å) of error matters for both designs:

Layer error	Triple well at 50 kV/cm	Triple well, field re-tuned	ADQW at 0 V	ADQW, Stark re-tuned
Outer / wide well ±3 Å	34–39%	34–39%	67–76%	69–79%
Barriers ±3 Å	45–48%	48%	62–63%	63–65%
Centre / narrow well ±3 Å	35–38%	48–53%	48–58%	85–92%

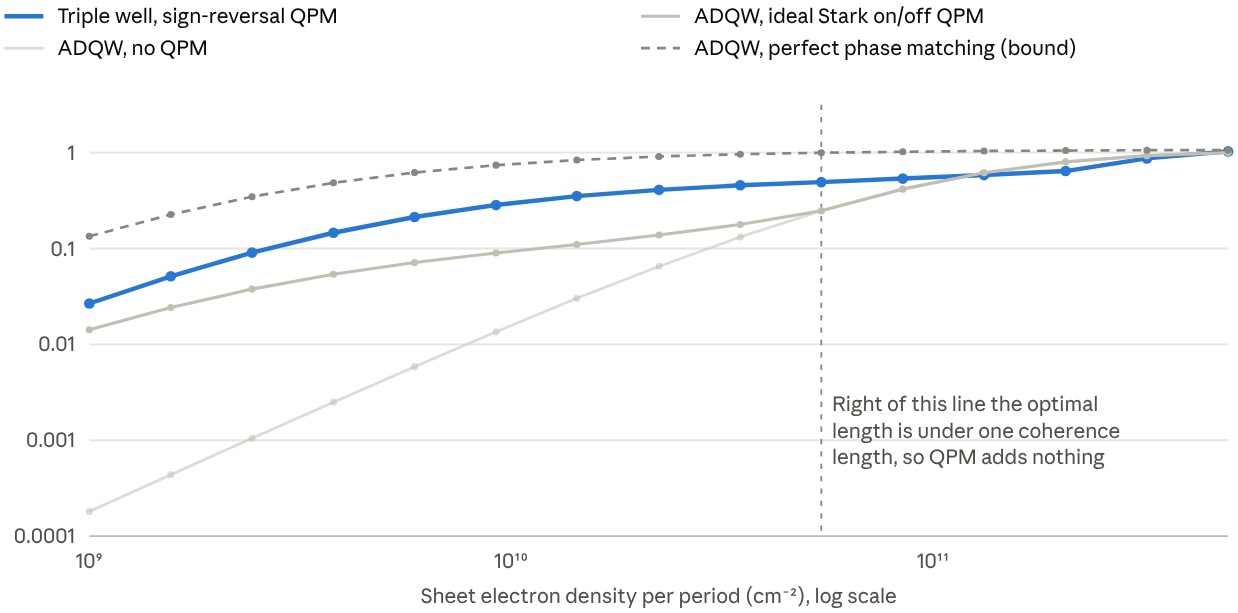
All values are % of the nominal ADQW. The outer-well width is the triple well's weak point, because it sets the doublet that the whole mechanism relies on. Re-tuning the pump to 9.45 μm recovers the worst case to 52%.

Device feasibility: where sign reversal pays off

Sign reversal only helps in long, low-loss waveguides. At the 5×10^{11} cm^{-2} doping used above, absorption limits the best device to 16–27 μm, shorter than the 63 μm coherence length, so no QPM scheme of any kind adds anything. Below about 5×10^{10} cm^{-2} the optimal length grows to hundreds of μm to millimetres, and there the triple well with sign reversal gives 1.9–3.2× the conversion of an ideally Stark-switched ADQW.

In long, lightly doped guides, sign reversal gives 2–3× a Stark-switched ADQW

Peak SH conversion per unit pump intensity at the optimal length; 1 = best ADQW without QPM at 5×10¹¹ cm⁻²



computed with qwchi2.py + propagate.py · Eq. 15.29, 300 K, 50% overlap, 3 cm⁻¹ background loss, coherence length 63 μm

The propagation model is the book's Eq. 15.29 with an undepleted pump below saturation, 50% modal overlap and 3 cm⁻¹ background loss at both wavelengths. Stark switching is treated as ideal, with no index grating, which flatters the ADQW. Even so, nothing beats the heavily doped ADQW in conversion *per unit pump intensity*. The case for N1 is therefore the low-doping regime: lower heat load per unit length for cw pumping, and the long nonlinear sections of THz difference-frequency sources (Sec. 15.5).

Quantity	Value	Comment
Coherence length, 10 → 5 μm SHG	63 μm (bulk InP); 62 μm with intersubband dispersion	Electrode period ≈ 126 μm; bulk index from the Pettit–Turner Sellmeier, valid to 10 μm
Bias, 30 periods (1.13 μm stack)	±5.7 V per segment at 50 kV/cm	Alternating top electrodes over a common bottom contact
Screening by the electrons' own dipole	≈7 kV/cm of the 50 kV/cm	Ground state shifts ≈41 Å; apply ≈57 kV/cm, or solve Schrödinger–Poisson

Quantity	Value	Comment
Tunnelling escape, levels 1–3	$< 10^{-20}$ meV	Negligible at this field
Thermionic leakage, 300 K	≈0.13 A/cm² (≈0.75 W/cm²)	Barrier ≈0.40 eV above the ground state
Index and absorption difference, + versus − segments	Zero by symmetry	No parasitic grating, unlike Stark QPM
Segment transition width	A few μm (cladding scale)	A few % loss of QPM amplitude, estimated, not modelled

Self-consistent re-check: Gate 2 passed

Including the electrons' own charge barely changes the answer. With self-consistent Hartree potentials and depolarisation shifts, the symmetric triple well gives 53–61% of the best ADQW across every model level and donor placement tested, against 55–56% before. Every material uncertainty and one-monolayer growth error tested keeps it above the 30% gate.

What was added. A periodic Schrödinger–Poisson loop for one doped period at fixed mean field: the stack is voltage-biased, so the electrons' screening redistributes the field inside the period but cannot change its average. Ando-type depolarisation shifts are applied to every populated transition. Both structures were re-optimised with the full model: the best ADQW is 64/10/29 Å (120.6 nm/V) and the best triple well is 58/12/37 Å at 70 kV/cm (66.3 nm/V, 55%). The 58/14/34 Å design stays the recommended one: 53% at only 55 kV/cm, so less voltage and leakage.

Model level, 58/14/34 Å vs 64/10/29 Å	Triple well (nm/V)	Field (kV/cm)	ADQW (nm/V)	Ratio
Fixed potential, no depolarisation (Gate 1 model)	64.3	50	115.0	56%
Fixed potential, depolarisation	62.1	45	101.8	61%

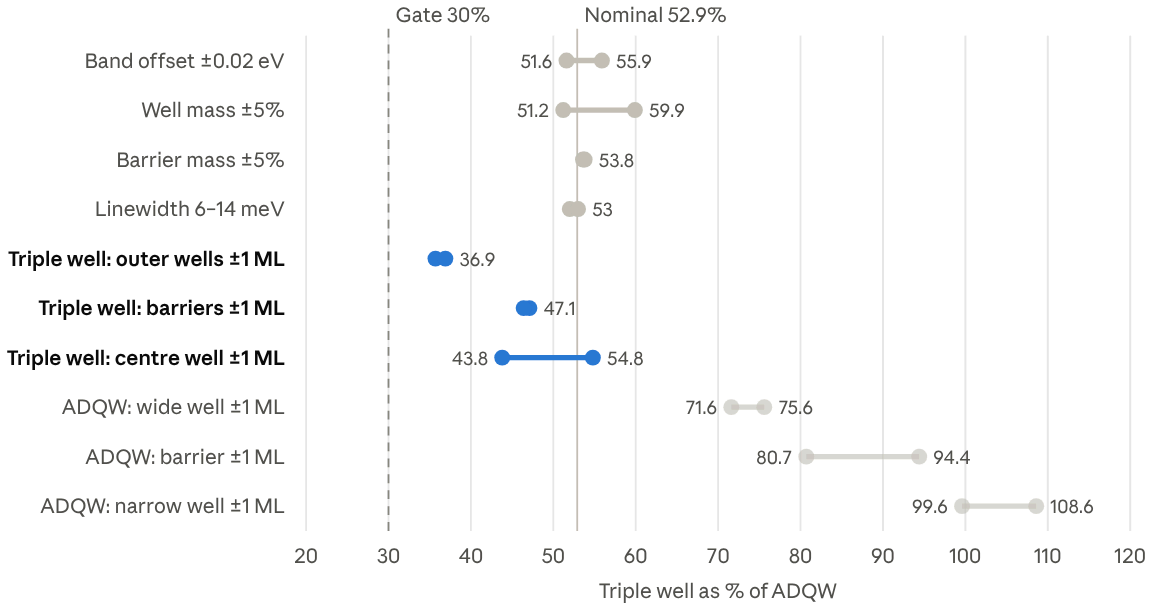
Model level, 58/14/34 Å vs 64/10/29 Å	Triple well (nm/V)	Field (kV/cm)	ADQW (nm/V)	Ratio
Self-consistent Hartree, no depolarisation	63.8	55	112.7	57%
Hartree + depolarisation (full)	**63.8**	**55**	**120.7**	**53%**
Full, donors in all wells	63.0	60	114.0	55%
Full, donors in the centre well	61.5	70	114.0	54%

The triple well's electrons move ~40 Å under bias, but screening only shifts the optimum field by about 5 kV/cm, because the redistribution happens inside a voltage-clamped period.

Every tested error keeps the triple well above the 30% gate

Triple-well |χ(2)| as % of the ADQW in the same case, full self-consistent model, field re-tuned; ADQW at 0 V

Blue: growth errors in the triple well. Grey: material uncertainty and ADQW growth errors.



computed with qwkit 0.2 (sp module) · studies/n1/errbudget.py · 300 K, 5×10¹¹ cm⁻² per period, donors at spacer centres

Two caveats from the error budget. The outer-well width remains the critical layer: ±1 monolayer takes the ratio to 36–37%, still above the gate but well below the 50% tie line. And the ADQW is compared at fixed 0 V, where its own growth errors hurt it more (72–

109%). A biased ADQW could partly re-tune, as the earlier non-self-consistent check showed, so the fair statement is that both designs need monolayer control, with the triple well's field acting as a built-in trim.

Next steps, all doable without a lab

The result is promising but rests on a non-self-consistent model. These steps either harden it into a publishable theory paper or kill it cheaply, in this order:

- [ ] **Self-consistent Schrödinger–Poisson with depolarisation.** *Done 10 Oct 2026 — passed; see the self-consistent section above.*
- [ ] **Saturation and rate equations.** Solve the full Eq. 15.29 with subband rate equations (book Sec. 15.4) at realistic intensities. The 15% thermal population of the second subband may also help as a reservoir; this is untested.
- [ ] **Detuned designs.** The book's Eq. 15.40 detuning lowers absorption faster than $\chi(2)$, which lengthens devices past one coherence length even at higher doping. Search detuned triple wells.
- [ ] **Waveguide and electrode electrostatics.** 2D mode solve for overlap and modal coherence length; electrostatic model of alternating top electrodes for transition width and fringing.
- [ ] **Robust-design search.** Re-optimise for tolerance to ±1 monolayer, not peak $\chi(2)$: outer-well sensitivity is the weak point.
- [ ] **Difference-frequency (THz) version.** Apply the same sign reversal to $\chi(2)(\Delta\omega; \omega_1, \omega_2)$ (Eqs. 15.44–15.46). A lasing QCL core cannot have its bias polarity alternated, so this needs a separate passive nonlinear section pumped by an external dual-wavelength QCL.

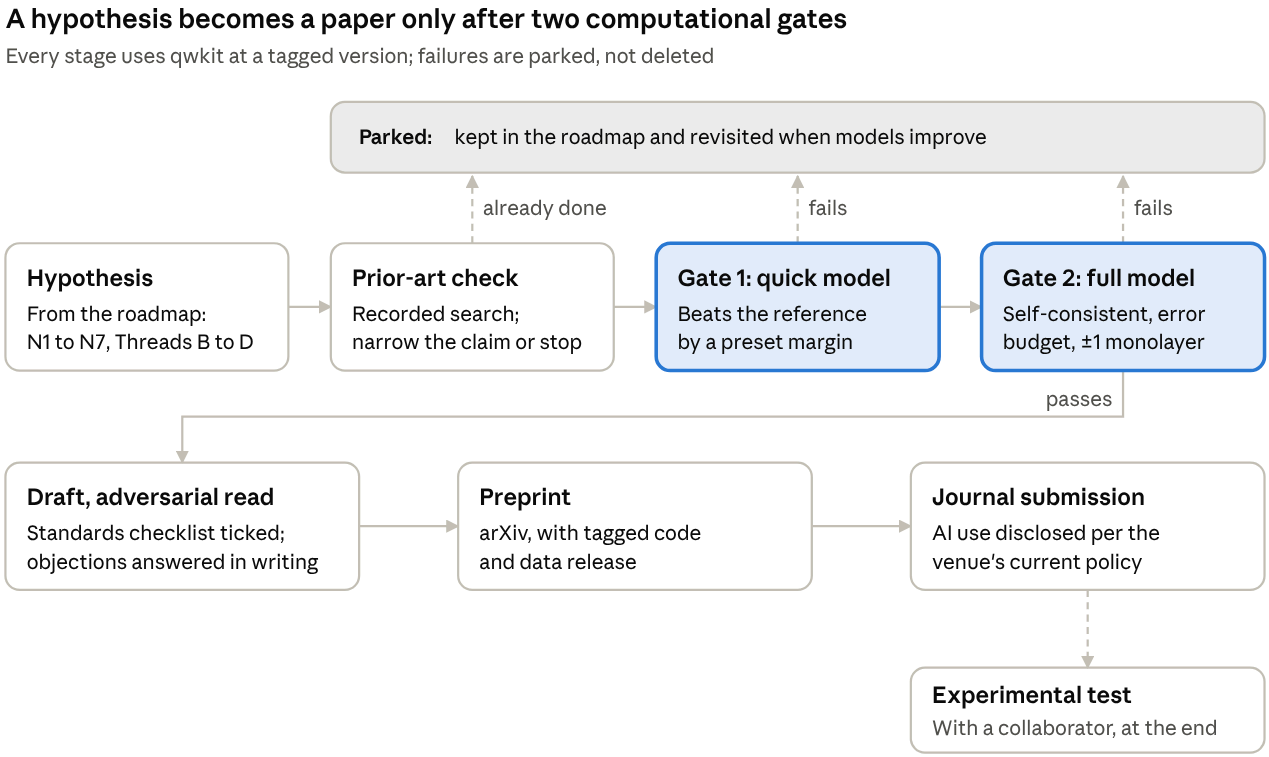
When to ask a grower. If the self-consistent and saturated models still give ≥40% of the ADQW and ≥2× a Stark-switched ADQW at a practical doping, the experiment is modest: one ~30-period InGaAs/InAlAs stack on InP, about 1.5 μm thick, plus lithographic top electrodes. Paik and Venkatesan work mainly on oxides. The closest III–V expertise to this chapter is Rui Q. Yang's interband cascade group and its antimonide MBE at the University of Oklahoma Quantum Device Laboratory, if that is your institution; Hinkey and Yang are in the chapter's suggested reading.

Publication framework

Oct 10, 2026 · how this programme turns calculations into papers

The programme publishes computational and theoretical papers, written so that an experimental group can test each prediction. Three things hold it together. First, one shared code base, **qwkit**, produces every number in every paper and must pass its validation suite before a figure is made. Second, a fixed claims discipline: prior-art search, validation, error budget and an adversarial read happen before anything is written up. Third, a staged pipeline: a roadmap hypothesis becomes a paper only after passing two computational gates. Papers are aimed at established optics and photovoltaics journals, with an open-source software paper on qwkit itself as a credibility anchor.

Paper pipeline



paper pipeline · 7 stages, 2 gates

Gate 1 is the quick, non-self-consistent model with a preset pass mark. Gate 2 repeats the claim with the full model, the error budget and the one-monolayer growth test. Anything that fails is parked with its reason, so a later model improvement can revive it.

Paper slate

Six papers come straight from the roadmap, plus a software paper. They are ordered by readiness. Venues are candidates to choose from, not commitments.

#	Working title	Central claim	Status	Main work remaining	Candidate venues
P1	Electrically reversible asymmetric quantum wells for sign-reversal quasi-phase matching	A mirror-symmetric triple well gives bias-polarity-set χ(2) sign at ~50% of an ADQW, with no parasitic index grating	Manuscript drafted 10 Oct 2026 (N1_paper.pdf); author checklist pending	Read two late-found 1995–96 prior-art papers; author details; journal template; public code DOI	Optics Letters; Applied Physics Letters; Physical Review Applied
P0	qwkit: an open, validated toolkit for intersubband optics in quantum wells	Reproducible, test-backed solver for χ(1), χ(2), self-consistent electrostatics	v0.3.0: core, self-consistent, waveguide, electrode and propagation modules, 17 tests	Documentation, examples reproducing published results, licence	Journal of Open Source Software
P2	Ratchet intermediate-band cells designed with the intersubband nonlinear toolkit (N2)	ADQW design rules map onto two-step absorption and ratchet depth	Not started	Detailed-balance cell model, grating-coupled sub-gap absorptivity	Progress in Photovoltaics; IEEE Journal of Photovoltaics
P3	Quasi-Fermi-level suppression versus escape time in quantum-well solar cells (Thread B)	A predictive model of well QFLS suppression, tested against published PL/EL data	Not started	Drift-diffusion with escape and capture rates; re-analysis of published data	Journal of Applied Physics; Progress in Photovoltaics
P4	Narrowband near-field TPV: phonon-polariton emitters and cascade absorbers (N4)	Polar emitters plus 0.1–0.15 eV cascade cells can convert 300–600 °C heat efficiently	Not started	Fluctuational electrodynamics, cascade-cell detailed balance	Physical Review Applied; ACS Photonics

#	Working title	Central claim	Status	Main work remaining	Candidate venues
P5	Self-current-matching quantum-well subcells (N7)	Stark feedback reduces spectral-mismatch losses in 2-terminal stacks	Not started	Drift-diffusion, annual-spectrum energy yield	IEEE Journal of Photovoltaics
P6	Reverse-QCL injectors as energy-selective contacts (N5)	QCL injector design meets hot-carrier contact specifications	Not started	NEGF transport (largest new build)	Physical Review Applied

P1 and P0 are written first, together: the software paper documents the code that produced P1's results, and P1 gives the software a showcase. P2 reuses the most code, so it is the natural third paper.

The shared code base: qwkit

Every published number comes from qwkit at a tagged version, through a script in that paper's study folder that writes its data to JSON. Figures are drawn from those files, never retyped. The validation suite runs before any figure is regenerated.

Module	Does	Validated against	Status
core	Effective-mass states with nonparabolicity; dipoles; $\chi(2)$ with Fermi–Dirac populations; absorption; tunnelling escape	Infinite-well energies and z12 (within 1–2%); book Eq. 15.28 three-level limit; exact $\chi(2)=0$ and oddness for symmetric wells; translation invariance; state parity	Done, 5 tests

Module	Does	Validated against	Status
sp	Self-consistent Hartree for a doped period at fixed mean field; Ando depolarisation shifts	Analytic potential of charged sheets (<1%); exact symmetry; charge neutrality and convergence; depolarisation blue shift vanishing at low density	Done, 4 tests
propagate	Book Eq. 15.29 with QPM patterns, background loss and overlap	Analytic undepleted-pump solution	Script exists; to be folded in with saturation and rate equations
detailed_balance	Solar and TPV cells: Shockley–Queisser, intermediate band, cascade TPV	Published SQ curve; book Fig. 15.3 and Fig. 15.13	Planned (P2, P4, P5)
nearfield	Fluctuational electrodynamics for planar emitter–cell stacks	Published near-field transfer spectra	Planned (P4)
transport	Drift-diffusion with well escape and capture; later NEGF	Published dark J–V and QFLS data	Planned (P3, P5, P6)

The package and N1 study scripts are in the attached qwkit.zip. A public repository (GitHub, with a Zenodo DOI per release) should exist before the first preprint, because reviewers and the software journal will ask for it.

Standards before anything goes in a manuscript

Each item is a box to tick per paper; a paper with an unticked box is not submitted.

- [ ] **Novelty.** A recorded search (Google Scholar, arXiv, Web of Science or Scopus through the library, patents) and a closest-prior-art table in the paper. Claims are narrowed to what the table shows is new — N1 already narrowed once this way.

- [ ] **Validation.** Every new quantity is reproduced against an analytic limit or a published number before it is used, and the test is added to the suite.
- [ ] **Error budget.** Results are re-run with conduction-band offset ±0.02 eV, effective masses ±5%, linewidth 6–14 meV, and ±1 monolayer layer errors. The headline claim must survive all of them.
- [ ] **Model ladder.** The headline number is shown at two levels of model (for example, non-self-consistent and self-consistent) so readers can see what each approximation costs.
- [ ] **Falsifiable prediction.** Each paper ends with a measurable prediction and the simplest experiment that would test it, with layer sequence, doping and expected signal.
- [ ] **Adversarial read.** Someone who did not do the work tries to break the central claim; objections and answers are kept with the study.
- [ ] **Reproducibility.** Tagged code version, data files, and one command per figure.
- [ ] **AI-use disclosure.** Journals forbid listing AI tools as authors and require disclosure of their use; authors remain responsible for every statement (AIP Publishing guidance). APS updated its journal AI policy in June 2026 (APS news); read the target journal's current policy before submitting, and keep a log of where AI assistance was used.

Paper 1 outline: electrically reversible asymmetric quantum wells

Working abstract. Quasi-phase matching of resonant intersubband $\chi(2)$ has so far relied on modulating its magnitude — by intermixing, or by Stark-detuning resonances — or on growth-defined orientation reversal. We show that a mirror-symmetric coupled triple well behaves as an asymmetric double well whose orientation is set by the sign of an applied field. Alternating-polarity electrodes therefore reverse the sign of $\chi(2)$ without changing linear absorption or refractive index, avoiding the parasitic grating of Stark-modulated schemes. Self-consistent calculations for InGaAs/InAlAs on InP give 53–61% of an optimised ADQW's $\chi(2)$ at 50–55 kV/cm, and at least 36% under band-offset, effective-mass, linewidth and single-monolayer growth errors. Propagation modelling identifies the lightly doped, long-waveguide regime where sign reversal gives 2–3× the conversion of Stark modulation.

Section	Content	Figure	Status
1 Introduction	Intersubband χ(2), the absorption/phase-matching problem (book Sec. 15.4–15.5), prior QPM schemes	Prior-art table	Drafted in N1 tab
2 Concept	Mirror symmetry ⇒ χ(2)(−F) = −χ(2)(F); identical linear response in ± segments; reversible-ADQW picture	Band diagrams at ±F and 0	Analysis done; figure to make
3 Model	Nonparabolic effective mass, density-matrix χ(2) with populations, self-consistent Hartree, depolarisation; validation suite	Validation table	Done, including self-consistent Hartree and depolarisation
4 Design and figure of merit	Symmetric single, double and triple wells vs optimised ADQW; the field-polarised doublet mechanism	χ(2) versus field (exists, non-self-consistent)	Numbers done; redraw figure with the full model
5 Robustness	Error budget: band offset, masses, linewidth, ±1 monolayer; field and pump re-tuning	Tornado chart	Done (range chart in N1 tab)
6 Propagation	Eq. 15.29 with QPM patterns; when QPM matters; doping regime	Conversion versus doping (exists)	Needs saturation and rate equations
7 Device proposal	Electrode layout, ±5.7 V segments, screening, leakage, transition regions; falsifiable prediction for a first sample	Device schematic	Waveguide mode and electrostatics to do

Section	Content	Figure	Status
8 Outlook	Detuned designs; THz difference-frequency version; GaN or Ge/SiGe platforms	—	Not started

The minimal publishable unit is Sections 1–5 plus a short Section 7, which would fit a letter-length paper. Sections 6–8 extend it to a full Physical Review Applied-style article.

Prior art and next papers

Oct 10, 2026 · @Gracen

The N1 idea itself (sign-reversing the field-induced χ(2) of symmetric wells with bias polarity) was patented in 1989 and 1994; what is new, and publishable, is the triple-well design that makes it competitive and the quantitative case for when it wins.

Prior-art verdict for N1

The concept is anticipated; the design and the quantitative assessment are not. The manuscript now concedes the concept, cites both patents, and claims only what is new.

Work	What it covers	Overlap with N1
Fejer & Byer, US 4,880,297 (1989)	Biased square wells whose χ(2) sign follows the field; spatially periodic bias in a waveguide, period an odd number of coherence lengths; claims speak of varying field *intensity*	Anticipates periodic-bias QPM in symmetric wells
Delacourt, Pocholle & Papuchon, US 5,289,309 (1994)	+χ(2) and −χ(2) zones written by field direction in a symmetric (stepped) well; no electrodes or waveguide analysis	Anticipates the sign-reversal concept
Tsang, Ahn & Chuang, APL 52, 697 (1988); Fejer et al., PRL 62, 1041 (1989)	Field-induced χ(2) in a single symmetric well, sign set by bias	Physics only, no QPM
Meyer et al., APL 67, 608 (1995)	Phase matching imposed electrically by voltage-tuning the resonant index	Different mechanism (true PM); full text not yet read
Vurgaftman et al., Solid State Commun. 100, 663 (1996)	Detuned, field-modulated resonances for broadband waveguide SHG	Possibly close; full text not yet read

Work	What it covers	Overlap with N1
Belkin et al., APL 88, 201108 (2006)	Stark on/off QPM along a QCL ridge	The magnitude-modulation benchmark

Still new: the mirror-symmetric coupled triple well that reaches 53% of an optimized ADQW at 55 kV/cm (single, double and the patent's stepped wells: 34–43% at 140–165 kV/cm); the linear-grating analysis; the self-consistent error budget; and the saturation-aware comparison against realistic benchmarks.

- [ ] Read Meyer 1995 and Vurgaftman 1996 (SSC) in full through the OU library before submission

What the revision found

The headline survives a harder test: with realistic relaxation and benchmarks, sign reversal wins by more at watt-level pumping, but absolute efficiencies are model-sensitive, so the paper now leads with ratios.

Check	Result	Effect on the paper
LO-phonon relaxation (new `scattering` module, benchmark within 13%)	Triple-well pump level drains into level 2 (0.79 ps), which empties in 2.6 ps; electrons pile up there	Absolute efficiency at 10 W falls about 3×; triple-well advantage grows
Realistic Stark on/off QPM (off segments with their own loss and index)	Up to 2.8× better than the old "idealized" on/off, which was not an upper bound	Benchmark replaced; old claim corrected
Head-to-head above about 1 W	1.9–4.7× realistic Stark; 1.0–1.9× ideal growth-reversed ADQW	New abstract numbers
Low power (10 mW)	0.63–1.03× Stark; 0.63–0.86× growth reversal	Stated plainly as the regime where N1 loses
Sensitivity (linewidth 6–14 meV, three relaxation models)	Ratios 1.15–2.3 and 1.3–3.4 hold; efficiency spans 0.09–0.9% at 10 W	New Table 4

Check	Result	Effect on the paper
Patent's stepped symmetric well	38% of ADQW at 160 kV/cm	Added to Fig. 2
Residual linear grating	None at the QPM wavevector; 30 cm^{-1} at 2K from the gaps vs 235 cm^{-1} at K for Stark	Claim made precise
Bias screening and heating	DC bias screened in about 0.6 μs at 300 K; 1.3 K/ns heating at 10 W	Pulsed bias and pump (≤10 ns) required
Solver audit	One RK4 sub-step at domain edges read the next domain's sign (0.3% error)	Fixed; all propagation reruns; 23/23 tests pass

The frontier, 2020–2026

The field has moved to metasurfaces and THz difference-frequency sources; waveguide QPM, saturation-aware design and many-body second-order theory are thin, which is where a solo theorist with qwkit fits.

Thread	Where it stands	Gap we could fill
Electrically tunable ISB metasurfaces	Bias tunes SH magnitude and phase (Yu et al., Nat. Photon. 2022); $\chi(2)$ sign still set by growth order (Sarma et al., Opt. Express 2022)	Bias-polarity sign control in metasurfaces using symmetric coupled wells
Programmable QPM in other platforms	SiN $\chi(2)$ written optically under 500–1000 V, 0.47 pm/V (Yanagimoto et al., Nature 2026)	ISB route gives $10^5\times$ larger $\chi(2)$ at ~30 V; reconfigurable-period designs

Thread	Where it stands	Gap we could fill
THz difference-frequency QCLs	Cherenkov phase matching shown not to be automatic (Oberhausen et al., APL Photonics 2023); metasurface THz sources saturate near 20 kW/cm² (Krakofsky et al., Nat. Photon. 2026)	Electrically tuned collinear QPM for DFG
Si-compatible ISB nonlinearity	Ge/SiGe waveguide SHG modelled at ~0.6 %/W, no QPM and no saturation (Chesi et al., Opt. Express 2023)	Port qwkit to Ge/SiGe; QPM and saturation for CMOS photonics
Saturation limits	Analytic limits exist for metasurfaces only; waveguide treatments neglect saturation	General saturation-limited efficiency bounds
Many-body second-order response	Multisubband-plasmon theory is linear; Ando shift is first order	Second-order RPA/TDHF for heavily doped coupled wells
Nitride and vdW wells	ScAlN/GaN telecom SPDC proposals (2026 preprint); gate-tunable WSe_2 ISB transitions (2025)	Polarization-field-assisted sign reversal at 1.5–3 μm

Next papers

Paper 2 should be the saturation-limit theory: it generalizes the strongest finding of N1, needs only code we already have, and is useful to every ISB nonlinear-optics group.

Rank	Paper	Calculation	qwkit reuse	Target venue	Effort
1	Saturation-limited efficiency of resonant ISB waveguides	Closed-form optimum doping, length and transition strength vs I_sat; validate with the LO-rate propagation; compare InGaAs, GaAs, Ge/SiGe	Full	Phys. Rev. Applied / JOSA B	2–3 months

Rank	Paper	Calculation	qwkit reuse	Target venue	Effort
2	Electrically programmable QPM for THz DFG	$\chi(2)(\omega_1-\omega_2)$ of symmetric wells under ±V; THz modes with phonon dispersion; bias-tuned period across 1–6 THz	High; waveguide needs a Lorentz-Drude extension	Optica / APL Photonics	3–4 months
3	Inverse design of bias-switchable heterostructures	Gradient or genetic search over layers with self-consistency in the loop; objective = sign contrast × I_sat with ±1 ML tolerance	Full	npj Comput. Mater. / ACS Photonics	2–3 months
4	Second-order RPA beyond the Ando shift	Coupled depolarization matrix for 3+ subbands; where single-shift theory fails for $\chi(2)$ magnitude and phase	Core + new theory	Phys. Rev. B	3–5 months
5	Ge/SiGe symmetric coupled wells for CMOS QPM	Port band parameters (L-valley, strain); repeat N1 for Si photonics	Medium	Opt. Express	2 months
6	Mid-IR photon pairs from bias-programmed QPM	Pair rate with resonant loss and thermal background; chirped bias for bandwidth	Medium	PRX Quantum / Optica Quantum	3 months
7	Bias-switchable ISB emitters in TPV detailed balance	Bias-dependent absorption → emissivity → TPV efficiency; bridges back to the solar/TPV half of the roadmap	Medium	Solar RRL / Phys. Rev. Applied	2 months

- [ ] Confirm paper 1 is not already covered by Gomez-Diaz et al. (PRB 92, 125429, 2015), which treats metasurfaces only
- [ ] Before paper 2, check Fujita/Belkin THz DFG work from 2024–2026 for electrically tuned QPM

Sources

- US 4,880,297 (Fejer & Byer)

- US 5,289,309 (Delacourt, Pocholle & Papuchon)
- Tsang, Ahn & Chuang, APL 52, 697 (1988)
- Belkin et al., APL 88, 201108 (2006)
- Vodopyanov et al., APL 72, 2654 (1998)
- Janz, Chatenoud & Normandin, Opt. Lett. 19, 622 (1994)
- Boucaud et al., Opt. Lett. 16, 199 (1991)
- Yu et al., Nat. Photon. 16, 72 (2022)
- Yanagimoto et al., Nature 649, 330 (2026)
- Chesi et al., Opt. Express 31, 17098 (2023)
- nextnano LO-phonon benchmark (Ferreira & Bastard reproduction)
- L. R. Ram-Mohan publication list

Frontier items without links above (Sarma 2022, Oberhausen 2023, Krakofsky 2026, nitride and WSe_2 work) come from a literature survey and are checked at title or abstract level only.

Research atlas: every open thread

Oct 10, 2026 · @Gracen

51 publishable paper ideas survive de-duplication across solar cells, TPV, nonlinear optics, modelling methods and new materials; 45 need no experiment. The ranking is driven by what qwkit already computes: intersubband (ISB) papers can start now, while most solar and TPV device papers first need one shared interband/escape module.

Progress

Paper	Status	Date
N1, sign-reversal QPM in symmetric triple wells	Revised 11 Oct 2026 after an equal-freedom comparison: the bias-optimized triple well beats growth reversal (1.5–7.9×) and Stark on/off (up to 2.4×), but an ordinary ADQW under bipolar ±60–100 kV/cm bias is ~2× better still (Stark shift across the pump reverses the phase of χ(2)). Updated 12 Oct 2026: Vurgaftman SSC 1996 read, it tunes the coherence length with a uniform field and never alternates polarity, so the bipolar pattern is not anticipated; Yu et al. (Sci. Adv. 2025) demonstrated bias-programmed χ(2) phase by Stark detuning in a metasurface; both now cited. Decision pending on reframing; Meyer APL 67, 608 (1995) still unread (the file supplied was APL 67, 757)	12 Oct 2026

Paper	Status	Date
MS-01, qwkit software paper	Draft complete (9 pages, venue-neutral); qwkit 0.5.0 release-ready with 30 tests, pip packaging, MIT license, CI, contributing guide; independent review done and fixed. JOSS needs six months of public history, so the venue is open (SoftwareX, Comput. Phys. Commun. or J. Comput. Electron.)	10 Oct 2026
NL-02, saturation-limited efficiency bounds	Manuscript complete (15 pages, NL02_paper.pdf), independently refereed twice and revised. Key results: $\|\chi(2)\|^2 = Q \cdot \alpha_\omega \cdot \alpha_{2\omega}$ with $Q \propto z_{23}^2/(\gamma_{12}\gamma_{13})$; optimum-length efficiency $e^{-2}(\hbar\Omega_{23})^2/4\gamma_{12}\gamma_{13}$; a rigorous bound for any saturation, length and QPM; no saturation ceiling in waveguides (confirmed by a full density matrix); detuning restores the unsaturated limit (3–6× in real structures at 4 W). Revised 12 Oct 2026 after reading Almogy's 1995 thesis: its symmetric-ladder optimum (Eq. 9.15), length-limited bound (Eq. 9.37) and numerical no-ceiling result are now credited; new generalized length bound places the optimum devices within 1.4–2.5× of the shortest possible length (18 pages)	12 Oct 2026

Paper	Status	Date
NL-05, collective (time-dependent Hartree) second-order response beyond Ando	Manuscript complete (19 pages, NL05_paper.pdf), independently refereed three rounds and revised; new qwkit.tdh module (frequency domain and real time, 34 tests). Key results: pathway factors $F_p = 1 - \kappa_p \Delta/(\tilde{E}-\hbar\omega-i\gamma)$ with energy substitution ≈ κ=0 and single-length effective medium ≈ κ=1; $\kappa_{exc} = c\cdot\sqrt{(L12/L23)}$ (0.81–0.85 double wells, 0.52 triple well); diagonal (permanent-dipole) Hartree terms oppose the enhancement; partial occupation of level 2 forms a multisubband-plasmon doublet that cancels it (ρ2→0 restores +36%); merit changes −32% to +7%; double counting costs up to 47%; benchmarks reproduce Załużny 1995 (2.10 vs 2.5) and Krieguer–Todorov 2024 (design shift to collective double resonance; 2.4 vs ~3); under saturation the static Hartree field of pump-transferred electrons dominates (frozen-shift model fails 3× at 2e12, holds within 9% at 5e11, the NL-02 density)	11 Oct 2026

Paper	Status	Date
MS-02 + MS-03, implicit-differentiation sensitivities, UQ and growth-tolerant design	Manuscript complete (17 pages, MS02_paper.pdf), refereed twice and revised; new qwkit.sens module (37 tests). Key results: Jacobian at ~1/10 of a forward solve, accuracy set by forward convergence; growth errors scatter resonances by ~1 linewidth; linearized UQ fails for resonant quantities (factors 0.56–1.43), doping 'dominance' is an artefact; mean merit −7–10% vs $\|\chi(2)\|$ −27–38%; landscape multimodal, best nominal optima detuned and fail on growth errors; window-weighted robust design +10–15% over the centred published design; bias trimming +31–121% for every design	11 Oct 2026
TPV-05, temperature limits of bias-switchable ISB thermal emitters	Manuscript complete (12 pages, TPV05_paper.pdf), refereed twice and revised; new qwkit.thermal module (42 tests). Calibrated on the Noda GaAs MQW (E12 130.5 vs 132 meV; contrast 2.4–3.0 vs ~3 at 470 K). InGaAs peak ISB absorption ÷4.7 and width ×2.8 from 300 to 1000 K (in-plane nonparabolicity, LO broadening); GaAs loses 42–71% of electrons to L valleys. Matched resonators keep emissivity 1; contrast bounded by ISB/parasitic loss (≤2.4 at 1000 K); generation leakage exceeds modulated power above 320–500 K (InGaAs) / 570–740 K (GaAs), recharging above a few kHz. TPV with 0.22 eV cells: resonant ISB emitter 15× below blackbody in power; selective emitters pay only with near-ideal cells. Next: MAT-01; GaN/AlN follow-up parked	12 Oct 2026

Where to focus

Six theory papers, about 18–24 months of solo work, run from the open tool to its limits, many-body physics, robust design, energy and finally a new oxide material grown at OU. None needs an experiment; the last two end in proposed measurements.

Order	ID	Paper	Why here	Months	Venue	OU hook
0	N1	Bias-polarity sign-reversal QPM with symmetric triple wells	Written; parked checks below; add NL-01 interface tolerance as a robustness section	0–2	Phys. Rev. Applied	—
1	MS-01	qwkit software paper	No open code chains depolarization → LO rates → $\chi(2)$ → saturation → waveguide; makes every later paper citable	0–3	JOSS	—
2	NL-02	Saturation-limited efficiency and contrast bounds for ISB waveguides and metasurfaces	The 2025 review still uses two-level Maxwell–Bloch bounds; direct sequel to N1's strongest finding	2–7	Phys. Rev. Applied / Optica	—
3	NL-05	Second-order (TDHF/RPA) response beyond the Ando shift	Tests whether the bounds survive collective effects at high doping	6–11	Phys. Rev. B	—
4	MS-02 + MS-03	Differentiable, growth-tolerance-aware inverse design, with parameter uncertainty	No autodiff Schrödinger–Poisson exists; ML surrogates ignore growth errors	9–15	npj Comput. Mater.	—
5	TPV-05	Temperature limits of bias-switchable ISB thermal emitters for low-temperature TPV	Noda's switchable emitter was shown near 300 K only; near-full code reuse	13–17	ACS Photonics	Rui Q. Yang (interband cascade cells)

Order	ID	Paper	Why here	Months	Venue	OU hook
6	MAT-01	ISB absorption and χ(2) at the polar $LaInO_3/BaSnO_3$ interface	Studied only for transport; polar asymmetry gives χ(2) with no composition steps	15–21	Phys. Rev. Applied / APL Materials	Hanjong Paik (MBE), T. Venkatesan (PLD)

First alternate: SC-02, hot-carrier thermalization in type-II InAs/AlAsSb wells, checked against the OU MBE group's published data (Santos, Mishima). **Second wave (from about month 18):** build one shared interband/escape module, ideally coupled to Solcore's interband absorption, which unlocks SC-01, SC-04, SC-05, SC-06 and TPV-01 together. **Time-sensitive:** NL-12 (realistic ISB polariton blockade) and NL-15 (switchable interband χ(2)) could be scooped by the originating groups.

Rui Q. Yang at OU works on interband-cascade TPV cells (about 9.6% for a 3-stage 0.4 eV cell; later 0.2 eV multistage devices), which is a closer match to this toolkit than thermal-radiation physics.

State of the art by area

The headline records (39.5% QW solar, 44% TPV) sit where qwkit reaches least; theory lags experiment most in ISB metasurfaces, switchable emitters and oxide or 2D wells, where qwkit reaches furthest.

Area	State of the art (2022–2026)	Main open problems
Nonlinear ISB optics	Metasurface $\chi(2)_{eff} \approx 55$ nm/V, SHG 0.14% at 1–10 W; electrically tuned THG with 1.1π phase (Park 2024) and full SH amplitude/phase control (Yu, Sci. Adv. 2025); CW THz DFG reported at CLEO 2024; ring-QCL solitons of ~1 ps at 8.3 μm (Nature 2025); interband coupled-well χ(2) of 2750 pm/V at 1550 nm (2026 preprint)	Multilevel saturation bounds (field still uses two-level formulas); many-body χ(2); interface tolerance; nonreciprocal time-modulated metasurfaces; realistic ISB polariton blockade

Area	State of the art (2022–2026)	Main open problems
QW solar cells	Strain-balanced GaInAs/GaAsP superlattices hold 39.5% (3J), 32.9% (2J), 34.2% AM0; 27.2% QW single junction; hot electrons at 490 K in OU's type-II InAs/AlAsSb wells; two-step upconversion at 5.1% vs a 46.8% limit	Unexplained ~10 mV Voc loss in thick superlattices; no ERE reported; reciprocity in MQW cells; realistic hot-carrier kinetics; field-optimized upconversion; radiation end-of-life
Thermophotovoltaics	41.1% tandem at 2400 °C, 38.8% at 1850 °C, ~44% air-bridge InGaAs at 1435 °C; near-field still below ~15%; no working thermophotonic device	QW absorbers in high-reflectance cells; QW thermophotonic LEDs; high-temperature behaviour of switchable ISB emitters; hot-carrier TPV with real cooling rates; interband-cascade TPV in near-field
Methods and software	Open codes cover pieces (kdotpy, Aestimo, ErwinJr2, Solcore, Sesame); NEGF with full scattering only commercial (nextnano); ML surrogates and Bayesian optimization tied to one reference design	No open ISB-optics chain; no differentiable solver; no uncertainty quantification of band parameters; no open multisubband-plasmon solver; no benchmark dataset
Emerging materials	$SrTiO_3/LaAlO_3$ ISB at room temperature (2018); $LaInO_3/BaSnO_3$ 2DEG with ~2100 $cm^2/V{\cdot}s$, transport only; gate-tuned ISB in few-layer WSe_2 (>100 meV) and InSe (>250 meV); ScAlN/GaN ISB at 515–709 meV; ZnO THz electroluminescence at 8.5 THz	Oxide ISB and $\chi(2)$ theory with real phonons and nonparabolicity; self-consistent gated 2D ISB and $\chi(2)$; Ga_2O_3 lifetimes; ScAlN polarization refit; Ge/SiGe switchable $\chi(2)$

Every paper opportunity

58 entries: 51 papers plus 7 parked directions (priority P4). Reuse: Full = existing qwkit, Ext = one or two new modules, New = a new solver class. Experiment: No, Opt (a proposed experiment written into the paper) or Collab (needs a partner's data). Effort: S ⩽ 3 months, M 3–6, L 6–12, XL > 12. Priority: P1 start within 12 months, P2 second wave, P3 opportunistic, P4 park. Every "not found" novelty call still needs a Scopus or Web of Science check.

Nonlinear intersubband optics

ID	Paper	Gap	Reuse	Expt	Venue	Effort	Priority
NL-01	Interface and monolayer tolerance of sign-reversal χ(2) designs	No tolerance study for engineered-sign stacks	Full	No	Fold into N1	S	P1
NL-02	Saturation-limited efficiency and contrast bounds, waveguides and metasurfaces	Only two-level bounds in use — done, see Progress	Full	No	PR Applied / Optica	M	P1
NL-03	Electrically programmable QPM for THz difference-frequency generation	No programmable-QPM THz waveguide design	Ext	Opt	PR Applied / APL Photonics	M	P2
NL-04	Mid-IR photon pairs from bias-programmed QPM	No ISB SPDC design found	Ext	No	Optica Quantum	M	P2
NL-05	Collective (TDHF/RPA) second-order response beyond Ando	Ando is a linear correction	Ext	No	PRB	M	P1
NL-06	Floquet Stark-modulated ISB metasurfaces for nonreciprocity	Proposed, never shown; RC limit 3.4 MHz	Ext	No	Nanophotonics	M–L	P2

ID	Paper	Gap	Reuse	Expt	Venue	Effort	Priority
NL-07	Non-perturbative ISB harmonic generation (THG to HHG)	No QW-specific HHG work found	New	No	PRB	L	P3
NL-08	Bandwidth and figure-of-merit bounds for ISB Stark modulators	10 Gbit/s devices, no general bound; now has a universal phase-per-loss bound $\Delta\varphi \leqslant (Q/2\gamma)\alpha L$ (literature review tab)	Full	No	PR Applied / IEEE JQE	S–M	P2
NL-09	Two-photon QWIP for picosecond QCL soliton autocorrelation	Solitons uncharacterized	Ext	Opt	PR Applied	M	P2
NL-10	Microscopic Kerr and linewidth-enhancement maps for ring-QCL solitons	Origin "not well understood"	Ext	No	PRB	L	P3
NL-11	Polariton–LO-phonon scattering design maps toward ISB polariton lasing	Rate "extremely low"; no design survey	Full	No	PR Applied	M	P2
NL-12	Materials-specific ISB polariton blockade (Oliveira–Ciuti mechanism)	Model-level proposal only	Ext	No	PRB / PRL	M	P2, time-sensitive
NL-13	Squeezed light from QCLs: cascade Langevin model	Excess noise blocks quantum QCL light	Ext	No	PR Applied	M	P3
NL-14	Sign-reversal QPM at near-IR in InAs/AlSb and GaN/AlN	Unexplored; GaN Isat 2.5× theory	Ext	No	Opt. Express	M	P2
NL-15	Bias-switchable interband $\chi(2)$ for electrically poled 1550→775 nm QPM	Existing designs fixed-asymmetric	New (k·p)	No	Optica	L	P2, time-sensitive

ID	Paper	Gap	Reuse	Expt	Venue	Effort	Priority
NL-16	Transmissive all-dielectric ISB metasurfaces	Needs full-wave solvers	New	No	Nanophotonics	L	P4
NL-17	Saturable-index-compensated (chirped) phase matching for detuned ISB conversion	Saturable resonant index costs 5–45% in detuned waveguides (NL-02)	Full	No	Opt. Lett. / Opt. Express	S	P1
NL-18	Absorption–nonlinearity identity and Rabi bound for ISB difference-frequency and THz generation	NL-02 covers SHG only; extend to the gain–nonlinearity identity for QCL intracavity DFG	Full	No	PR Applied / Optica	S–M	P1
NL-19	Depopulation-engineered coupled wells that remove the saturation bottleneck	Triple well loses up to 3× to an intermediate-state bottleneck (NL-02)	Full	No	IEEE JQE / PR Applied	M	P1
NL-20	Detuned-ladder SHG active regions at a fixed 10 μm pump (anharmonicity ±100 meV)	Detuning rule (NL-02) needs structures not yet designed; feeds MS-02	Full	No	Opt. Express	M	P1
NL-21	Coherent-drive ISB SHG beyond the rate-equation regime (multilevel density matrix)	NL-02 checked three levels only	Ext	No	PRB	M	P2
NL-22	Time-domain Maxwell–Bloch ISB propagation: sub-T1 pulses beyond steady-state saturation limits	NL-02 bounds are steady-state; experiments are pulsed	Ext	No	PRB / Optica	M	P1

ID	Paper	Gap	Reuse	Expt	Venue	Effort	Priority
NL-23	Strong-field symmetry breaking in symmetric triple wells: optically written χ(2) sign gratings	Predicted in general (Levinson et al. 1994), never designed for QPM	Ext	Opt	PR Applied	M–L	P2
NL-24	Saturation-engineered ladders: maximize the high-power slope, or a v2-dipole SHG power limiter	Closed-form C∞ (NL-02) not yet used for design	Full	No	Opt. Express	S–M	P2
NL-25	Loss budget of resonant ISB ridge waveguides (roughness, free carriers, metal)	Background loss sets the waveguide high-power ceiling (NL-02)	Ext	Opt	IEEE JQE	M	P2
NL-26	Identity-based bound on ISB sum-frequency upconversion detectors	No efficiency-per-absorption bound for upconversion detection	Full	No	Opt. Lett.	S	P2
NL-27	Saturation with population-dependent Hartree energies: charge-transfer feedback, moving-resonance detuning and possible SHG bistability in asymmetric ladders	Frozen-shift rate models miss a 3× effect at 2e12 (NL-05)	Full	No	PRB	M	P1
NL-28	k-resolved TDH/TDLDA second-order response with in-plane nonparabolicity and exchange-correlation	NL-05 is Hartree-only with a single subband dispersion	Full	No	PRB	L	P2

ID	Paper	Gap	Reuse	Expt	Venue	Effort	Priority
NL-29	Collective-coupling-aware design rules for high-density SHG ladders (κ engineering, avoiding level-2 occupation)	No design rule accounts for multisubband-plasmon mixing in doubly resonant ladders	Full	No	Opt. Express	M	P2
NL-30	Pathway-resolved collective factors in resonator-coupled QWs (beyond single-length ENZ models)	Metasurface ENZ models assume κ = 1	Full	No	ACS Photonics	M	P3
NL-31	Collective effects in electrically programmable χ(2) metasurfaces: do equal Stark coefficients survive depolarization, and can one meta-atom cover 2π?	Yu et al. 2025 and Park et al. 2024 design with bare transition energies at 1.5×10^{12} cm^{-2}; scoping shows the collective χ(2)(V) trajectory winds around the origin instead of tracing a cardioid	Full	No	Phys. Rev. Applied	M	P2
NL-32	Cardioid-engineered triple wells (equal Stark coefficients) for bipolar Stark QPM in waveguides	Gives an exact π phase difference at equal |χ(2)| for ±V; extends N1′s bipolar ADQW result	Full	No	Opt. Express	S	P2

NL-17 to NL-21 came out of NL-02. NL-02 also bears on N1: N1′s resonant 10 W operating point lies beyond the conservative rate-equation limit (about 4 W), and tuning the pump 25–35 meV below resonance raises its efficiencies 3–6×, so N1′s comparison between QPM schemes is worth repeating with a detuned pump before submission (added to the parked list).

NL-27 to NL-30 came out of NL-05. NL-05 also bears on NL-02: at the NL-02 density (5×10^{11} cm^{-2}) the frozen-shift saturation model holds to within 9%, partly through cancellation above 3 MW/cm^2; at higher density the static Hartree field of pump-transferred electrons must be included (NL-27).

NL-31, NL-32 and TPV-10 came out of the 12 Oct 2026 literature review (Almogy thesis, Vurgaftman SSC 1996, Yu et al. 2025). The same review adds a benchmark for MAT-01: Almogy's coupled-well modulator, where measured charge transfer per volt exceeded Hartree theory by 15% because of exchange.

Methods and software

ID	Paper	Gap	Reuse	Expt	Venue	Effort	Priority
MS-01	qwkit: an open, validated ISB-optics chain	No open code with this chain	Full	No	JOSS / CPC	S	P1
MS-02	Differentiable, fluctuation-aware inverse design	No autodiff Schrödinger–Poisson	Ext (JAX)	No	npj Comput. Mater.	M–L	P1
MS-03	Uncertainty quantification of QW observables vs band parameters	No UQ study found	Full	No	JAP	S–M	P1
MS-04	Cross-code benchmark and open heterostructure dataset	None exists	Full	No	J. Comput. Electron. / Sci. Data	M	P2
MS-05	Open multisubband-plasmon solver with phonon coupling	Theory exists, no open code	Ext	No	PRB / JOSS	M	P2
MS-06	Unified non-Lorentzian ISB lineshapes	Fragmented old models	Ext	No	JAP	M	P3
MS-07	Reduced-order electron–electron scattering for density-matrix models	Codes stop at Hartree	Ext	No	J. Comput. Electron.	M	P3
MS-08	Spurious-solution-free 8/14-band k·p benchmark	Ad hoc handling	New	No	J. Comput. Electron.	L	P3
MS-09	Continuum vs atomistic models for InGaN QWs	Accuracy map missing	New	No	PRB	L	P4
MS-10	Open QCL NEGF code	Heavy incumbents	New	No	—	XL	P4 (avoid)

Quantum-well solar cells

ID	Paper	Gap	Reuse	Expt	Venue	Effort	Priority
SC-01	Escape-to-device model for 80–300-period superlattices	NREL's ~10 mV Voc loss unexplained	Ext + New	No (Collab upside)	JAP / IEEE JPV	L	P2
SC-02	Hot-carrier thermalization in type-II InAs/AlAsSb QWs	Only generic type-I models	Ext	No (OU MBE upside)	PR Applied / JAP	M–L	P1
SC-03	Hot-carrier MQW limit with subband-designed selective contacts	2024 device gave 0.28 V	Ext	No	EPJ PV	M	P2
SC-04	Reciprocity and ERE breakdown in MQW cells	Treated for bulk only	Ext	No	PR Applied	M–L	P2
SC-05	Field-dependent two-step photon upconversion with TM intraband step	Optimal field open; 5.1% vs 46.8%	Ext	No	JAP	M	P2
SC-06	QW intermediate-band and ratchet limits with realistic ISB absorption	Limits use step absorptivity	Ext	No	PR Applied	M	P2
SC-07	Radiation end-of-life modelling of QW space cells	60% vs 72% retention unexplained	New	No	IEEE JPV	M	P3
SC-08	Inverse design of strain-balanced stacks for current matching	None found	New	No	Solar RRL	M	P3
SC-09	Luminescent coupling in GaInP/GaAs-QW tandems	No QW-specific study	New	No	IEEE JPV	S–M	P3
SC-10	Review and open dataset after the 39.5% record	No post-2022 review	—	No	Prog. Photovolt.	M	P3
SC-11	Valley-photovoltaic heterointerface theory	OU device limited by valley mismatch	New	Opt	PR Applied	L	P3
SC-12	2D Ruddlesden–Popper perovskites as MQW absorbers	SCAPS-only device work	New	No	Solar RRL	M	P4

Thermophotovoltaics

ID	Paper	Gap	Reuse	Expt	Venue	Effort	Priority
TPV-01	QW absorbers in air-bridge, high-reflectance TPV	No QW re-analysis since 2018	Ext + New	No	PR Applied	M–L	P2
TPV-02	Fluctuational electrodynamics + transport for QW near-field cells	Bulk cells only	New	No	PR Applied	L	P3
TPV-03	Interband-cascade TPV: near-field, thermoradiative, negative luminescence	Not modelled	New	Collab (R. Q. Yang)	PR Applied	L	P2
TPV-04	QW vs double-heterostructure thermophotonic LEDs	Requirements paper omits QWs	Ext + New	No	PR Applied	M–L	P2
TPV-05	High-temperature limits of switchable ISB thermal emitters	Shown near 300 K only	Full	No (Opt)	ACS Photonics	M	P1
TPV-06	Hot-carrier TPV with real QW cooling rates	Idealized cooling only	Ext	No	PR Applied	S–M	P2
TPV-07	What limits out-of-band reflectance above 99%	No general design rules	Ext	No	SEMSC	S–M	P3
TPV-08	Zero-gap spacer TPV with QW cells	Concept done	New	No	IJHMT	M	P4
TPV-09	Tutorial: quantum structures in thermal-photonic conversion	No QW-focused review	—	No	JAP Tutorial	M	P3
TPV-10	ISB emissivity switching by intra-period charge transfer between coupled wells instead of depletion	TPV-05 found depletion switches lose to generation leakage above 320–740 K; Almogy's 1995 coupled-well modulator moves charge without a depletion region	Full	No	J. Appl. Phys.	S	P3

Emerging material platforms

ID	Paper	Gap	Reuse	Expt	Venue	Effort	Priority
MAT-01	ISB absorption, χ(2) and lifetimes at $LaInO_3/BaSnO_3$ interfaces	Transport only; χ(2) at abstract level	Ext	Opt (Paik, Venkatesan)	PR Applied / APL Materials	M	P1
MAT-02	$SrTiO_3/LaAlO_3$ MQW ISB with multi-orbital masses and ε(E,T)	No temperature-resolved theory	Ext	Opt (Venkatesan)	PRB	M	P2
MAT-03	Self-consistent ISB and χ(2) in gated few-layer InSe and WSe_2	Infinite-well fits only	Ext	No	PRB / 2D Mater.	M	P2
MAT-04	β-Ga_2O_3 wells: polar-phonon lifetimes, χ(2), laser feasibility	Energies done (2020), lifetimes not	Ext	No	JAP	M	P2
MAT-05	ScAlN/GaN polarization refit and robust designs	Polarization exceeds theory	Ext	No	APL / JAP	M	P3
MAT-06	m-plane ZnO/MgZnO THz QCL design at 8–9 THz	Emission shown, no laser	Ext	No	PR Applied	L	P3
MAT-07	CdSe nanoplatelet ISB vs thickness: nonparabolicity benchmark	Rigorous fit unverified	Ext	No	J. Phys. Chem. C	S	P2
MAT-08	Ge/SiGe symmetric coupled wells with bias-switched χ(2)	No switchable design	New (scattering)	No	PR Applied	M–L	P2
MAT-09	GeSn/SiGeSn ISB parameter-sensitivity map	Uncertain band ordering	Ext	No	JAP	M	P3
MAT-10	$SrVO_3/SrTiO_3$ metallic-QW ENZ and plasmons	Optics unexplored	New	Opt	ACS Photonics	M	P4

ID	Paper	Gap	Reuse	Expt	Venue	Effort	Priority
MAT-11	$KTaO_3$, EuO, HgTe, deep-UV AlGaN, InAs/GaSb superlattices	Need k·p or DFT	New	—	—	XL	P4 (avoid)

Checks before starting each paper

Each programme paper has one go/no-go check; run it in Scopus or Web of Science before the start date, not after.

Paper	Check	Go if
MS-01	Overlap with Aestimo, ErwinJr2, kdotpy, Solcore; JOSS scope	The depolarization → χ(2) → waveguide chain is absent elsewhere
NL-02	Forward citations of Krakofsky et al. 2025 review and Mann et al. Optica 2021; arXiv 2403.15911	No multilevel, nonparabolic saturation-bound paper exists
NL-05	Rosencher/Bois-era many-body χ(2); citations of Cominotti PRB 2023 and the 2023 JAP plasmon perspective	No second-order TDHF/RPA ISB treatment exists
MS-02	Citations of the 2020 Bayesian-NEGF, 2024 MLP and 2026 ANN papers; "differentiable Schrödinger–Poisson", "JAX heterostructure"	No autodiff heterostructure optimizer is published
TPV-05	Noda-group follow-ups 2015–2026; GaN/AlGaN ISB emitters; "intersubband" + "thermophotovoltaic"	No theory of high-temperature ISB emitter linewidth and contrast
MAT-01	Demkov (UT Austin) and Belgrade $BaSnO_3$ nonlinear work 2022–2026; PARADIM stannate superlattices; $BaSnO_3/LaInO_3$ phonon energies and permittivities	No peer-reviewed $BaSnO_3$-QW χ(2) paper with nonparabolicity and phonons, and Paik confirms the structures are growable

Paper	Check	Go if
SC-02	Ferry (ASU) and Cavassilas (Marseille) InAs/AlAsSb models; citations of Makhfudz 2022; thermalization coefficients from OU papers	No type-II kinetic model with real band structure

- [ ] Before the second wave: decide whether to couple to Solcore's interband absorption or write a hole-subband module

Parked for a later stage

These N1 pre-submission steps are saved for later and do not block research. They are also in `LATER_STAGE.md` in the project folder.

- [ ] Read Meyer, Hoffman, Bartoli & Ram-Mohan, "Intersubband second-harmonic generation with voltage-controlled phase matching", APL 67, 608–610 (1995) in full (the file supplied on 12 Oct was their APL 67, 757 laser paper); Vurgaftman et al., SSC 100, 663 read 12 Oct 2026
- [x] ~~Confirm the second inventor's first name on US 5,289,309 (Jean-Claude or Jean-Paul Pocholle)~~
- [ ] Deposit the code on GitHub and Zenodo for a DOI
- [ ] Ask a colleague who did not do the work to attack the central claim
- [x] ~~Read J. Yu, J. Kim, H. Chung, J. Son, G. Boehm, M. A. Belkin & J. Lee, "Full complex amplitude control of second-harmonic generation via electrically tunable intersubband polaritonic metasurfaces", Sci. Adv. 11, eadw8852 (2025), doi:10.1126/sciadv.adw8852 (open access) in full: local metasurface phase control or propagation QPM? Cite the companion THG paper, Light Sci. Appl. 13 (2024), doi:10.1038/s41377-024-01517-y~~
- [x] ~~Before submitting N1: repeat its QPM-scheme comparison with a red-detuned pump (NL-02 shows 3–6× higher efficiency and operation inside the rate-equation regime)~~
- [x] ~~Before submitting NL-02: read Almogy & Yariv, Opt. Lett. 19, 1828 (1994) and Vurgaftman et al., IEEE JQE 32, 1334 (1996) in full; confirm neither already states the identity or the waveguide no-ceiling result~~
- [x] ~~Get the correct Vurgaftman, Meyer, Ram-Mohan paper (IEEE JQE 32(8), 1334–1346, Aug 1996) — done 11 Oct 2026; compared in NL-02 and the literature review tab~~
- [x] ~~Read Almogy & Yariv, J. Nonlinear Opt. Phys. Mater. 4, 401 (1995) and Almogy's 1995 Caltech thesis: the likeliest prior forms of NL-02's identity or limits~~

Sources

The full research report, with about 120 linked sources and the reasoning behind every row, is `docs/Quantum well research thread atlas.md` in the project folder. Key sources:

- Krakofsky et al., ISB polaritonic metasurface review, Nanophotonics 2025
- Park et al., electrically tunable ISB THG, Light Sci. Appl. 2024
- Kazakov et al., bright solitons in ring QCLs, Nature 2025
- Oliveira & Ciuti, ISB polariton quantum nonlinearities, arXiv 2025
- Ramesh et al., coupled-well interband χ(2), arXiv 2026
- France et al., strain-balanced QW cells, NREL 2022
- Esmaielpour et al., hot electrons in InAs/AlAsSb QWs, Sci. Rep. 2018
- Watanabe et al., two-step photon upconversion, Kobe 2021
- LaPotin et al., 41% TPV, 2022
- Papadakis et al., thin near-field TPV cells, 2021
- Inoue et al., switchable ISB thermal emitter, Nat. Mater. 2014
- OU interband-cascade TPV, OSTI
- Correa Hernandez & Gmachl, ML QCL design, 2024
- Solcore
- Lee, Schlom, Char et al., $LaInO_3/BaSnO_3$ 2DEG, 2024
- Ortmann et al., $SrTiO_3/LaAlO_3$ ISB, ACS Nano 2018
- Shcherbakov et al., gate-tuned ISB in InSe, 2024
- Lyman & Krishnamoorthy, Ga_2O_3 ISB theory, 2020

Literature review: Almogy & Yariv (LEOS 1994) and Vurgaftman et al. (IEEE JQE 1996)

Oct 10, 2026 · @Gracen

Almogy and Yariv's 1994 summary partly anticipates NL-02 and independently confirms it: our density-matrix solver reproduces their closed-form saturated susceptibility to five digits, and our ceiling condition gives their μ12/μ23 = √2. The uploaded Vurgaftman 1996 paper is not the one NL-02 cites (that is IEEE JQE 32, 1334, August 1996), but it still opens three threads. The review adds eight research threads and four items to read before NL-02 is submitted.

What each paper does

Paper	What it does	Key result	Relevance to us
G. Almogy, A. Yariv, "Quantum well intersubband nonlinear optics: beyond perturbation theory," Proc. LEOS'94, vol. 1, pp. 179–180 (invited NO3.1)	Rotating-wave, non-perturbative solution of the optical Bloch equations for a dc-biased two-level well (electro-optic effect) and a three-level ladder driven by an intense pump (SHG)	EO: the Stark shift moves the Lorentzian exactly, so the EO coefficient saturates and changes sign and the index change is capped by the transition's own index. SHG: $\chi_{eff} = \chi_{res} / [1 + 2\Omega^2 T_2^2 (1 + 3T_1/T_2)]$, $\Omega = \mu E/2\hbar$; SHG saturates at about half the absorption saturation intensity; resonance is optimal below saturation and detuning above it "if propagation length limitations are removed"; SHG vanishes asymptotically for $\mu_{12}/\mu_{23} = \sqrt{2}$. Efficiency plotted for one absorption length, peaking near 1% with I_{SH} saturating at $I_{sat}/16$	High: closest prior art to NL-02 Secs. 5–6; an exact benchmark for our solvers; points to strong-field SHG in symmetric wells (Levinson et al., PRA 1994) and to ISB birefringence (Almogy et al., APL 1993)

Paper	What it does	Key result	Relevance to us
I. Vurgaftman, P. N. Freeman, P. K. Bhattacharya, J. Singh, "Time-domain optical wave propagation in disordered and nonuniform guiding structures," IEEE JQE 32, 2095–2104 (Dec 1996)	Implicit, locally one-dimensional finite-difference solver of the 2-D scalar wave equation for pulses in waveguides with roughness and longitudinal nonuniformity	Etched-mirror reflectivity falls as $\exp(-a \cdot L \cdot \delta/\lambda^2)$, $a \approx 10$, for roughness depth δ and correlation length L; a lateral waveguide couples light into an absorbing HBT collector within about 10 μm	Low for NL-02 directly. Useful for: background loss from sidewall roughness (which sets NL-02′s high-power ceiling), segment boundaries in electrode-written QPM (N1), pulsed time-domain propagation, and waveguide-to-absorber coupling

Checks against our work

Every statement of Almogy and Yariv we could test agrees with NL-02 once their relaxation convention (each excited level decays straight to the ground state at 1/T1) is used. Script: `studies/nl02/ay_check.py`.

Their statement	Our result	Verdict
$\chi_{eff}/\chi_{res} = 1/[1 + 2\Omega^2 T2^2(1 + 3T1/T2)]$	Full density matrix: identical to 5 digits for T1/T2 = 5–50 and ΩT2 = 0.03–3. Rate equations: $1/(1 + 6\Omega^2 T1T2)$, i.e. the same without the coherent term $2\Omega^2 T2^2$	Exact agreement; explains why our rate equations are accurate at resonance (error below T2/3T1, about 2% for T1/T2 = 15)
SHG vanishes asymptotically for $\mu12/\mu23 = \sqrt{2}$	Our closed-form ceiling condition r23 = (t32+t31)/(1+t31) gives $\sqrt{2}$ for decay to ground; density matrix: $\chi \cdot I$ falls as $1/I$ there, constant for equal dipoles	Exact agreement

Their statement	Our result	Verdict
Detuning preferable above saturation if propagation length is not limited	Quantitative rule (detune pump by γ√(I/Is), harmonic oppositely to keep α2ω = 2αω) recovers Theorem 1; 15–31× in the ladder, 3–6× in qwkit structures	Qualitative prior art; NL-02 now credits it
Efficiency for one absorption length peaks near 1%, I_SH saturates (I_sat/16)	With the length optimized there is no ceiling (Sec. 5); a fixed length behaves like our thin film	Consistent; their ceiling is a fixed-length effect
EO: index change capped by the transition's own index	Our identity analogue for Stark phase modulators: phase per neper of loss ⩽ Q/2γ, exactly, independent of doping, dipole and length (Q = Stark shift)	New bound, consistent with their cap and with Almogy's thesis preferring absorption modulation

What changed in NL-02

- Introduction and Discussion now credit Almogy and Yariv's non-perturbative treatment, their fixed-length efficiency, their qualitative detuning remark, and their limits papers (J. Nonlinear Opt. Phys. Mater. 4, 401, 1995; Caltech thesis, 1995); also Khurgin (JOSA B 1989), Almogy, Segev and Yariv on index-induced phase matching (Opt. Lett. 1994) and Goldoni's χ(2) quantum limit (J. Appl. Phys. 2001).
- Sec. 2 gives the analytic reason the rate equations work at resonance: the exact solution differs only by the coherent term $2\Omega^2T2^2$, a relative error below T2/3T1. Our solver reproduces the exact form to five digits.
- Sec. 5 notes that our ceiling condition reproduces their √2.
- Sec. 6 opens by crediting their detuning remark and states that Theorem 1 makes it quantitative.
- The novelty that remains: the identity $|\chi(2)|^2 = Q\cdot\alpha\omega\cdot\alpha2\omega$; Theorem 1 at the optimum length; Theorem 2 for any saturation, length and QPM; the no-ceiling result with the length optimized and its closed-form slope; the υ = 1 detuning rule; the thin-film contrast; and the qwkit validation. Whether the 1995 review or thesis already contains the identity or the no-ceiling result is the main open risk.

New research threads

ID	Thread	Origin	Expt	Priority
NL-08 (upgraded)	Universal phase-per-loss bound for ISB Stark phase modulators: $\Delta\varphi \leqslant (Q/2\gamma)\cdot\alpha L$, so a π shift costs $\geqslant 2\pi\gamma/Q$ nepers (13.6 dB at $Q = 2\gamma$); add bandwidth and RC limits	Almogy–Yariv EO result + NL-02 identity; verified numerically	No	P1
NL-18 (broadened)	Gain–nonlinearity identity for intracavity DFG in QCLs: the same inverted transitions give gain and χ(2), so	χ(2)	$^2 \propto$ gain product × (THz dipole)^2/ γ^2, bounding THz conversion per mid-IR power	NL-02 identity applied to inversion
NL-22	Time-domain Maxwell–Bloch ISB propagation: sub-T1 pulses avoid population saturation, so pulsed SHG may exceed the steady-state limits; new qwkit module	Vurgaftman et al. time-domain method + NL-02 steady-state theorems	No	P1

ID	Thread	Origin	Expt	Priority
NL-23	Strong-field symmetry breaking in N1′s symmetric triple wells: optically written χ(2) sign gratings and even harmonics without bias	Levinson, Segev, Almogy, Yariv, PRA 49, R661 (1994)	Opt	P2
NL-24	Saturation-engineered ladders: choose z23/z12 and the relaxation paths (LO-phonon placement) to maximize the high-power slope s∞, or set z12/z23 = √2 for an SHG power limiter	NL-02 closed-form C∞ + Almogy–Yariv zero	No	P2
NL-25	Loss budget of resonant ISB ridge waveguides (sidewall roughness, free carriers, metal): background loss sets NL-02′s high-power ceiling	Vurgaftman et al. roughness model + NL-02	Opt	P2

ID	Thread	Origin	Expt	Priority
NL-26	Bound on sum-frequency upconversion detectors (mid-IR to near-IR) per absorbed pump, from the identity; waveguide-coupled absorber design	NL-02 identity + Vurgaftman HBT coupling example	No	P2
N1-b	Segment-boundary scattering in electrode-written QPM: confirm that sign reversal has no index step, unlike Stark on/off	Vurgaftman et al. nonuniform-waveguide method	No	Fold into N1

NL-17 (chirped or power-matched phase matching) also gains support: Almogy's thesis identified the ISB index as a possible dominant limit on conversion, and NL-02's density-matrix check finds it is the main correction under detuning.

Vurgaftman, Meyer, Ram-Mohan, IEEE JQE 32, 1334 (1996)

The correct paper (library copy, read 11 Oct 2026) solves numerically the same rate-equation and coupled-wave model as NL-02, with the population-dependent resonant index, heating and waveguide modes; it does not contain the identity, either theorem, or the no-ceiling result. It is now credited in NL-02 (17 pages) in five places.

Point in the paper	Bearing on NL-02
Eq. 5: $\eta_s = (\hbar/\Gamma\tau_{12})(z_{13}z_{23}/z_{12}^2)^2(\delta_{12}^2+\Gamma^2)^2/[(\delta_{13}^2+\Gamma^2)\Gamma^2]$, "lowest-order estimate at saturation", independent of doping, angle, index and photon energy	Source of the textbook formula; credited for the doping independence. Our identity adds that z12, z13 and detuning also drop out at fixed intensity once harmonic absorption is included

Point in the paper	Bearing on NL-02
Eqs. 3–4: dimensionless propagation with coupling $\sqrt{\eta s}$ and $s = \gamma\omega\alpha^0 L$	Same structure as our ladder model (their $\eta s = 4\times$ our coupling², the field-convention factor)
Fig. 8 and Conclusion: double resonance best below saturation, detuning above; length held at the overall optimum	Second qualitative precedent for NL-02 Sec. 6; their falling curves are the fixed-length ceiling
Predicted 14–16% at 20–40 MW/cm² (InGaAs/InAlAs waveguide) with z23/z12 = 3.2 (8-band)	qwkit gives the same energies (within 3 meV) but z23/z12 = 1.1; the TRK sum rule allows 3.2 only if z12 ⩽ 0.8 nm; the 2–3 Rabi energy at their operating point is 1.1–2.9×
Includes the resonant index along the device ($\cos \int \Delta n\, ds$)	Captures the saturable-index correction our density matrix found; credited

The cross-check also exposed a trap in qwkit: core.dipoles on raw nonparabolic states gave origin-dependent dipoles. It now orthonormalises by default; a mirror-symmetry regression test was added (31 tests pass). Published N1 and NL-02 numbers were unaffected, because their pipelines already orthonormalised.

New thread from this paper: MS-04, multiband (8-band k·p) dipoles for ISB nonlinear design, quantifying how much one-band models misestimate z23/z12 near anticrossings.

Still to read before submission

- ~~Vurgaftman, Meyer, Ram-Mohan, "Optimized second-harmonic generation in asymmetric double quantum wells," IEEE JQE 32(8), 1334–1346 (Aug 1996), doi 10.1109/3.511546 — read 11 Oct 2026 (library copy), see the section below~~
- ~~Almogy, Yariv, "Resonantly-enhanced nonlinear optics of intersubband transitions," J. Nonlinear Opt. Phys. Mater. 4, 401–458 (1995), doi 10.1142/S0218863595000173: most likely place for prior forms of the identity or limits~~
- ~~Almogy, Caltech PhD thesis (1995), doi 10.7907/3qj9-9777: contains "a limit on the second-harmonic intensity in a given propagation length" (scanned PDF)~~
- ~~Almogy, Yariv, "Second-harmonic generation in absorptive media," Opt. Lett. 19, 1828 (1994)~~

Sources

- Almogy & Yariv, LEOS'94 (doi 10.1109/LEOS.1994.586950)
- Vurgaftman, Freeman, Bhattacharya, Singh, IEEE JQE 32(12), 2095–2104 (1996), uploaded copy (DOI not verified)
- Almogy thesis record, Caltech
- Levinson et al., PRA 49, R661 (1994)
- Almogy, Shakouri, Yariv, APL 63, 2720 (1993)
- Almogy & Yariv, JNOPM 4, 401 (1995)
- Goldoni, J. Appl. Phys. 89, 1755 (2001)
- Cominotti et al., PRB 107, 115431 (2023)
- Crossref record of the LEOS'94 paper

Załużny, Phys. Rev. B 51, 9757 (1995) and Phys. Rev. B 47, 3995 (1993)

Read in full from the library copies supplied on 11 October 2026.

PRB 51, 9757 (1995): depolarization and SHG in asymmetric step wells. Second-order RPA (density matrix, relaxation-time approximation) for a three-level GaAs step well, with the microscopic Coulomb matrix elements L(i,j;k,l), which are our S_ij,kl. Rotating-wave approximation, only the ground subband occupied, only near-resonant terms kept.

- Energy substitution is his "diagonal approximation".
- The 1→2 / 2→3 coupling multiplies $\chi(2)$ by $c(\omega) = (\bar{E}21 - \hbar\omega - i\Gamma)/(\bar{E}21 - \hbar\omega - i\Gamma)$. This is exactly the NL-05 excited-state factor F_exc, with $\kappa_{exc} = \bar{\alpha}32/\alpha22$.
- In his example (GaAs, 10^{12} cm^{-2}) the coupling raises max $|\chi(2)|^2$ by 2.5×. Two-level wells: depolarization only shifts the SHG peak.
- No permanent-dipole (diagonal) Hartree terms, no thermal populations, no multisubband mixing in propagation, no efficiency.

Consequences for NL-05 (done): credited throughout; new benchmark reproducing his step well. Optimum step height and photon energy agree to within 1.2 meV; enhancement 2.10 vs his 2.5. The diagonal terms he omitted lower it to 1.87.

PRB 47, 3995 (1993): nonlinear ISB absorption with depolarization. Two-level rectangular well in RWA. The depolarization shift falls with intensity and the saturated line becomes asymmetric, so energy substitution with a fixed shift fails under saturation. He notes that

the static field of optically generated space charge also shifts the subbands but is small in rectangular wells. In NL-05's asymmetric double well this static charge-transfer field is the dominant saturation correction (Sec. 8).

Citation correction: the earlier "Bondarenko & Załużny, J. Phys.: Condens. Matter 12, 8267 (2000)" was wrong (that page range is an unrelated 2001 paper) and has been removed. An OSTI/ETDEWEB record with the title "Effects of the Coulomb interaction on intersubband doubling and difference-frequency mixing in asymmetric double quantum wells" exists; its journal is still unidentified.

Almogy thesis, Vurgaftman SSC 1996 and Yu et al. 2025 (read 12 Oct 2026)

Almogy's thesis already contains the symmetric-ladder case of NL-02's Theorem 1, a length-limited bound and the no-ceiling observation; NL-02 now credits all three and generalizes the bound. The thesis covers the JNOPM 4, 401 material (Chs. 5–9), so the JNOPM paper and Opt. Lett. 19, 1828 are ticked above.

Source	What it contains	Effect on our work
Almogy, Caltech PhD thesis (1995), Ch. 9	Optimum length at $\alpha_2\omega = 2\alpha\omega$ (Eqs. 9.11–9.13); for equal dipoles, detunings and widths the amplitude conversion is $\Omega T_2/e$, independent of dipole size, density and detuning (Eq. 9.15); length-limited bound on the harmonic intensity (Eq. 9.37); 25% needs ~100 absorption lengths and >100 I_s on resonance (Fig. 9.6); opposite-sign detuning pays above saturation (Figs. 9.7–9.9)	NL-02 intro, Theorem 1, Secs. 4–6 and discussion rewritten to credit these. New `length_bound.py` reproduces Eq. 9.37 exactly and generalizes it: optimum devices above 3 I_s sit at 1.4–2.5× the minimum length

Source	What it contains	Effect on our work
Almogy thesis, Ch. 10	Coupled-well charge-transfer modulator, 50 periods GaAs/$Al_{0.4}Ga_{0.6}As$; transfer per volt 15% above Hartree (exchange); blue shift and broadening with population	New benchmark for MAT-01; new thread TPV-10 (switching without depletion)
Vurgaftman, Meyer, Ram-Mohan, SSC 100, 663 (1996)	64.5/18/36.5 Å InGaAs/InAlAs ADQW waveguide: 14% at 10.6 μm and 20 MW/cm²; a uniform −25 kV/cm switches the SH off by shortening the coherence length; ≥1% from 9 to 20 μm with the optimum field	No polarity reversal and no sign change of χ(2): N1′s bipolar pattern is not anticipated. N1 text corrected; full title "… in a waveguide" fixed in both bibliographies
Yu et al., Sci. Adv. 11, eadw8852 (2025)	Triple well with equal Stark coefficients: χ(2)(V) traces a cardioid; two flipped meta-atoms give 0–2π phase and on/off amplitude; electrically programmed SH gratings; Stark slope 3.37 meV/V	Local phase grating, not propagation QPM; cited in N1 as experimental support for detuning-based phase control. New threads NL-31 and NL-32

Two supplied files were other papers: Meyer et al., APL 67, 757 (1995), a type-II laser paper, instead of APL 67, 608; and Zhou et al., Light Sci. Appl. 13, 242 (2024) instead of the THG paper, which is Park, Yu, Boehm, Belkin & Lee, Light Sci. Appl. 13, 169 (2024), doi 10.1038/s41377-024-01517-y.

Full research report behind the Research atlas tab · Gracen Ownby, University of Oklahoma · 10 October 2026

Intersubband physics anchors a solo quantum-well research programme

About seventy raw ideas from the five topic notes and the earlier frontier survey reduce to **58 de-duplicated entries: 51 actionable papers and 7 parked or avoided directions**. **45 of the 51 need no new experiment.** What separates them is less novelty than toolkit reach. qwkit v0.4 already runs the full conduction-band intersubband (ISB) chain. So nonlinear-optics, ISB-emitter and oxide/2D intersubband papers can start now. Nearly every solar-cell and thermophotovoltaic (TPV) device paper, by contrast, first needs valence-band states, interband absorption and escape or transport modules that qwkit lacks.

The strongest papers that can be written now are six: - a qwkit software paper; - saturation-limited efficiency and contrast bounds for ISB nonlinear devices, going beyond the two-level Maxwell–Bloch formulas the field still uses (Krakofsky et al. 2025); - a collective (TDHF/RPA) second-order response beyond Ando; - differentiable, fluctuation-aware inverse design of bias-switchable stacks (no automatic-differentiation heterostructure solver turned up in the notes' searches); - temperature limits of electrically switchable ISB thermal emitters for low-temperature TPV; - ISB absorption and $\chi(2)$ at the polar LaInO3/BaSnO3 interface, which has been studied only for transport (Lee et al. 2024) and sits next to Hanjong Paik's oxide MBE.

Together these make an 18–24-month programme. It runs from tool, to limits, to many-body physics, to design, to energy, to new materials, and ends with proposed experiments at OU. The most valuable energy threads should come in a second wave, after one shared interband module exists. These are: - hot-carrier thermalisation in type-II InAs/AlAsSb wells, checked against published OU data; - a device model for 80–300-period strain-balanced superlattices that explains NREL's unexplained ~10 mV Voc loss (France et al. 2022); - reciprocity breakdown in multiple-quantum-well (MQW) cells.

Before N1 is submitted, two items must be read and cited: - **Yu et al., Sci. Adv. 11, eadw8852 (2025)**, on full complex-amplitude electrical control of ISB SHG in polaritonic metasurfaces. Only its metadata was retrieved, and it is the closest prior art to bias-polarity sign reversal (Nature Index record). - **Park et al., Light Sci. Appl. 13, 169 (2024)**, which showed electrically induced phase gratings in ISB third-harmonic generation (Park et al. 2024).

Every "not found" novelty call rests on roughly 30–55 web-search calls per area. Confirm each one in Scopus or Web of Science before writing.

qwkit reaches every intersubband question but few interband ones

The reuse column in the master table follows from what qwkit v0.4 contains: - 1D Schrödinger–Poisson with Kane nonparabolicity - Löwdin orthonormalisation - Ando depolarisation - Fröhlich LO-phonon rates - density-matrix $\chi(2)$ - multilevel saturation rate equations - a planar TM waveguide solver - an electrode Laplace solver - coupled-mode SHG with quasi-phase matching (QPM) and segmented media

All of it is parameterised for InGaAs/InAlAs.

The reuse labels mean: - **"Full":** the paper needs only these modules plus a new material parameter set. - **"Extension":** the paper adds one or two modules inside the same 1D framework. Examples are hole subbands, transfer-matrix tunnelling, thermal subband populations, polarisation sheet charges, multi-mode Fröhlich sums, a layer-chain Hamiltonian, or a JAX port. - **"New code":** the paper needs a new solver class, such as multiband $k\cdot p$, drift-diffusion, fluctuational electrodynamics, real-time TDSP, RCWA/FDTD or NEGF.

This split explains most of the ranking below. The nonlinear, ISB-emitter, polariton and emerging-material ISB papers score "Full" or light "Extension". The solar and TPV device papers need hole states, interband absorption with excitonic edges, escape/tunnelling rates and some drift-diffusion.

That gap is not a reason to abandon the energy threads. It is a reason to build one interband/escape module once and reuse it. The same module serves: - the superlattice device model (SC-01); - reciprocity breakdown (SC-04); - two-step photon upconversion (SC-05); - QW absorbers in air-bridge TPV (TPV-01); - thermophotonic QW LEDs (TPV-04).

There is also a cheaper route. Solcore already gives Python 8-band k·p and Chuang-method QW absorption with TE/TM matrix elements and excitons. Its stated weakness is the absence of open boundaries, so unbound states come out wrong (Solcore). Coupling qwkit's escape physics to Solcore's interband absorption, rather than rewriting k·p, is the effort-minimising path. It also forbids framing any paper as "a new QW solar-cell solver".

Nonlinear intersubband optics: experiments now outrun a two-level theory

State of the art

Polaritonic ISB metasurfaces are the most active experimental front. A 2025 review puts MQW $\chi(2)$ and $\chi(3)$ **3–5 orders of magnitude above natural crystals**, with ISB dipoles of about 100 D. The benchmark results it cites are: - **$\chi(2)$_eff ≈ 55 nm/V** (Lee et al., Nature 2014); - **0.14% SHG power efficiency** at 1–10 W pump; - Kerr $\chi(3)$ above 10^{-13} m²/V² with sub-2 ps response; - saturable reflection dropping from ~0.64 to below 0.1 between ~100 kW/cm² and ~10 MW/cm².

The same review still uses a simplified two-level Maxwell–Bloch model. Its stated bounds are: - power-limiter contrast $\Delta R \le 1/(1 + 2\gamma_a\gamma_2/\Omega^2)^2$; - a maximum Rabi frequency of **$\Omega_{max} \approx 0.26\ \omega_q$**, against the 0.13 ω_q used to date; - a warning that saturation already matters at ~50 kW/cm², so "pump, material and photonics must be co-optimized" (Krakofsky et al. 2025).

Electrical control has advanced quickly. Park et al. reported: - **$|\chi(3)_{zzzz}| = 7.15\times10^{-14}$ m²/V²** at 9 μm; - **450% modulation depth**, **~1.1π phase tuning** and ±10.7° beam steering in ISB THG; - a four-subband Poisson–Schrödinger model, whose predicted Stark tuning overshot the measurement (attributed to nonuniform Schottky band bending); - RC-limited bandwidth of only **~3.38 MHz** (Park et al. 2024).

The 2025 Science Advances paper extends this to full complex-amplitude SHG control (Nature Index record).

The review calls THz DFG from ISB metasurfaces "only theoretical". That conflicts with a CLEO 2024 report of **continuous-wave THz DFG** that beats photomixers at 2.5–5 THz. Its reported $\chi(2) > 10^5$ pm/V has a garbled exponent and should be treated with caution (Krakofsky et al., CLEO 2024).

Time-dependent Schrödinger–Poisson work predicts an intensity-driven redshift, because the dipole shrinks and the depolarisation shift is suppressed. That coherent nonlinearity is distinct from incoherent saturation (Cominotti et al., PRB 2023).

Adjacent fronts

Quantum cascade lasers (QCLs). Monolithic ring QCLs now host: - free-running dark solitons (Opačak et al., Nature 2024); - DC-driven bright solitons of **~1 ps at 8.3 μm** (Kazakov et al., Nature 2025).

Their microscopic inputs are still poorly understood: - The Bloch-gain Kerr coefficient (~10^{-15} m²/W) comes with a dispersive lineshape whose origin is "not well understood" (Opačak et al., PRL 2021). - NEGF gives linewidth-enhancement factors of 0.1–1. Two-level models reach only ~0.04, and density-matrix approximations can get the sign wrong at the gain peak (Franckié et al.).

ISB polaritons and quantum ISB optics. ISB polaritons now show: - stimulated coherent scattering (Knorr et al., PRL 2022); - spontaneous polariton–LO-phonon scattering, though at an "extremely low" rate (Goulain et al. 2025).

Quantum ISB optics remains theory. The leading proposal is photon-exchange-bound bound-to-continuum polaritons with Rydberg-like giant nonlinearities (Oliveira & Ciuti 2025).

Interband $\chi(2)$ and modulators. On the interband side: - Asymmetric AlGaAs/GaAs coupled wells reached **2750 pm/V** at 1550 → 775 nm, more than 7× bulk GaAs. Growth interruptions pushed them toward ideal-interface predictions (Ramesh et al. 2026). - A coupled-well metasurface reached **~14 nm/V effective** (Fathi et al. 2026). - Unipolar Stark modulators carry **10 Gbit/s at 9**

µm (Dely et al.). - GaN/AlN ISB switches measured Isat ≈ 9.6 W/µm² against a 3.9 W/µm² theory. A claimed 30× reduction in coupled wells was never demonstrated (Li et al. 2007).

Open problems

Four problems stand out: - rigorous multilevel bounds for conversion efficiency and switching contrast; - many-body second-order response at high doping; - time-modulated, nonreciprocal metasurfaces, which have been proposed but not demonstrated (the closest analogue is a superconducting JoFET metasurface, Taravati 2024); - interface-tolerance analysis for engineered χ(2).

Where N1 sits

Bias-polarity sign-reversal QPM itself was anticipated by US 4,880,297 (1989) and US 5,289,309 (1994). N1's novelty is narrower: - **the design:** the mirror-symmetric triple well, which reaches 53% of an optimised asymmetric double quantum well (ADQW) at 55 kV/cm; - **the comparison:** saturation-aware, giving 1.9–4.7× over realistic Stark QPM and 1.0–1.9× over an ideal growth-reversed ADQW above ~1 W.

The Science Advances 2025 paper controls the SHG phase locally in a metasurface rather than through propagation QPM. That distinction must be verified from the full text and stated explicitly in N1. Park et al. 2024 must be cited as the precedent for electrically written phase gratings.

Quantum-well solar cells: 39.5% records, but no validated device model

Records and the unexplained losses

Strain-balanced GaInAs/GaAsP superlattices with up to 300 periods hold the one-sun records: - **39.5% (3J)** and **32.9% (2J)**, with **34.2% AM0** (NREL); - a **27.2% certified** GaAs-QW single junction at Voc 1.040 V (Steiner et al. 2021).

France et al. thinned the barriers from 170 Å GaAsP to 20 Å GaP and found three things they could not fully explain (France et al. 2022): - tunnelling and thermionic lifetimes become equivalent at **~5 nm** barrier thickness; - thickening the GaInAs from 850 to 2550 nm cut the gap by 15 meV but Voc by 25 mV, and the extra **~10 mV is "not identified"**; - whether GaP-barrier stacks are optically thick without a rear reflector is "not known".

Neither NREL paper reports external radiative efficiency (ERE) or per-well absorption. Temperature- and suns-dependent work on the same devices concludes the field must "move beyond simple drift-diffusion models" (Ahmed et al., JAP 2025).

Radiation hardness is a clean modelling target (Hubbard, SPW 2023). After 1 MeV electrons at 10^{15} cm^{-2}, QW single junctions: - keep 92% of Jsc against 81% for controls; - lose Voc from 0.996 to 0.760 V; - retain only **60% of efficiency against 72%**.

Next-generation concepts remain at proof of principle

The OU-led hot-carrier line is the most relevant here: - Type-II InAs/AlAsSb wells reached a steady-state hot-electron temperature of **490 K**, with thermalisation stretched "from picoseconds to nanoseconds" (Esmaielpour et al., Sci. Rep. 2018). - Thicker barriers tune the phonon bottleneck (Esmaielpour et al., APL 2021). - Valley photovoltaics, done at OU, did not exceed the single-gap limit. The cause was valley-degeneracy mismatch at the heterointerface (Esmaielpour et al., Nature Energy 2020).

The theory behind these is still generic. Makhfudz et al. model type-I wells with simple bands and find a non-monotonic thickness dependence once hot LO phonons and screening are included (Makhfudz et al. 2022). A 2025 review notes "some debate" on thermalisation pathways (Afshari et al. 2025). An InGaAs/AlInAs single-QW energy-selective contact gave Voc 0.28 V without hot-carrier operation, and its authors say the confined level must be redesigned (Boyer-Richard et al. 2024).

Two-step photon upconversion (TPU) cells reach **5.1%** against a 46.8% detailed-balance limit. The Kobe group lists the optimal interface field (16.6 kV/cm gave a ~45% IR-induced EQE boost) and the tunnelling/extraction balance as open (Watanabe et al. 2021). Intermediate-band absorption is known to depend strongly on band filling, more than two orders of magnitude for E_L (Ramiro et al., PRB 2017). Yet intermediate-band and ratchet limits still use idealised step absorptivities (Zhang & Krich 2020).

Records elsewhere do not use QWs

As far as the notes found, the leading cells outside all-III–V multijunctions do not use QWs: - the Fraunhofer III–V/Si triple junction at **36.1%** (Fraunhofer ISE); - perovskite/Si at **35.2%** (pv magazine 2026).

Pure 2D Ruddlesden–Popper "quantum-well" perovskites sit near 20–21%, and are modelled mostly with SCAPS (RSC 2026). One unresolved point: efficiency tables v67 appeared in Prog. Photovolt. (ISFH), but pv magazine reports v68 "in Joule". Verify the venue before citing.

Implications for this researcher

The **highest-fit solar paper is hot-carrier kinetics in type-II InAs/AlAsSb**. It reuses the nonparabolic SP and Fröhlich modules, it is checked against published OU data, and it connects directly to Santos and Mishima's MBE.

The **highest-impact solar paper is the thick-superlattice device model**, where NREL and the Sellers group have published the unexplained numbers. It needs the most new code and faces direct competition from the experimental groups, so it belongs in the second wave. Sellers now appears in University at Buffalo listings and co-authors with NREL's France. That is an inference from listings, not a confirmed arrangement.

Thermophotovoltaics: 44% records, with quantum structures largely absent

Far-field records and what limits them

Far-field TPV has reached: - **41.1%** with a 1.4/1.2 eV tandem at ~2400 °C and 2.39 W/cm² (LaPotin et al. 2022); - **38.8%** with 0.74 eV GaInAs at 1850 °C and 3.78 W/cm² (Tervo et al. 2022); - **~44%** (43.8% for 0.90 eV InGaAs at 1435 °C) with a near-99%-reflective air-bridge cell (pv magazine 2024).

The limits are out-of-band reflectance, parasitic free-carrier absorption, series resistance and emitter durability. The first calorimetric Ge TPV measurement (7.3%) identified free-carrier absorption in the doped substrate as the dominant loss (Datas group 2026). Commercial pull is real: - Fourth Power plans a 25 MW / 250 MWh graphite unit and a 1 MWh pilot (MIT News 2026); - Antora claims >40% cells from a 2 MW/yr line (Antora).

Near-field TPV and thermophotonics

Near-field TPV is still below ~15% efficiency (Jeyar et al. 2024). NREL scaled it to a 0.28 cm² pair at a 150 nm gap, giving ~25× the far-field power (Selvidge et al. 2024). A zero-vacuum-gap spacer doubles far-field power density but calls for cells below 0.7 eV (Habibi et al., EES 2025).

Thermophotonics (LED + PV) still has no functional device. GaAs efficiency falls from 63% to 18% as EQE drops from 100% to 95%, and the requirements analysis does not discuss QW active regions (Sadi et al.).

QW and ISB structures are nearly dormant in TPV

QW-TPV has been almost dormant since 2018. The live quantum-structure threads are very-low-gap cells: - OU's interband-cascade TPV: a 3-stage ~0.4 eV cell gave **9.6%** at 300 K (pre-window, ~2017) (OSTI), with later ~0.2 eV multistage devices (OU QDL); - Montpellier's 0.23 eV InAs/InAsSb barrier cells: Voc ~75 mV at 273 K, no efficiency reported (pv magazine 2026).

Theory favours thin absorbers. Photon recycling lets cells thin down and shed Auger volume, and a thin InAs near-field cell exceeds 50% at a 1100 K emitter (Papadakis et al. 2021). Coupled fluctuational-electrodynamics/drift-diffusion models exist for bulk cells only (Callahan et al. 2021).

On the emitter side, Noda's carrier-depletion ISB emitter switched emissivity from **0.74 to 0.24**, about 10^4× faster than thermal modulation (Inoue et al., Nat. Mater. 2014). No theory of how such emitters behave at 500–1000 K turned up.

Hot-carrier TPV has an idealised detailed-balance treatment claiming near-Carnot efficiency near maximum power (Nimje et al. 2023), but realistic QW cooling physics has not been added.

The opening for this researcher

The open niche is the intersection the field skipped: quantum-structure physics in modern TPV regimes. The emitter paper (TPV-05) is the only TPV entry that reuses qwkit almost entirely. It also creates a natural pairing with Rui Q. Yang's interband-cascade cells, whose mid-IR band edges match ISB emission.

Methods and software: open codes are fragmented, and none is differentiable

The open-code landscape

The open ecosystem covers pieces but not the ISB-optics chain: - **kdotpy** covers lattice k·p and was published in SciPost Phys. Codebases (kdotpy). - **Aestimo 1D** covers 1D Schrödinger–Poisson and was published in Comput. Mater. Sci. 2021 (Aestimo). - **ErwinJr2** covers QCL design (PyPI). - **Solcore** and **Sesame** cover solar cells (Sesame). - **NEGF** with LO/acoustic phonons, interface roughness, alloy disorder and e–e scattering exists commercially in nextnano (nextnano.NEGF).

No open, tested chain combining depolarisation, LO-phonon rates, $\chi(2)$, saturation and waveguides was found. That is the qwkit software paper's justification.

Design optimisation and machine learning

Design work has moved quickly: - **Bayesian optimisation with NEGF** helped reach the 210 K THz-QCL record. The same study found period thickness must be held within a few percent (PR Applied 2020). - **An MLP surrogate** screened ~907 million designs in 8 h and raised a figure of merit from 94.7 to 141.2 eV ps Å² (Correa Hernandez & Gmachl 2024). - **A 2026 ANN surrogate** trained on 62,250 density-matrix samples gave a ~90,000× speed-up (Stojković et al. 2026).

Every surrogate is local to one reference design and ignores growth fluctuations. The only "inverse design" heterostructure hits were 1996–1999 inverse-scattering papers (cond-mat/9903445). No differentiable Schrödinger–Poisson solver turned up.

Unsettled band-structure methods

Several basics remain unsettled: - Spurious-solution handling is still ad hoc. A 2023 8-band InAs/GaSb study pays "careful attention" to it without giving a general recipe (Murphy et al. 2023). - No uncertainty-quantification study propagating band-parameter spreads into QW observables was found. - Self-consistent GW inside NEGF has been demonstrated only for carbon nanotubes (Deuschle et al. 2023).

Many-body ISB optics

The multisubband-plasmon framework is established theory with a 2023 perspective (Montes Bajo et al., JAP 2023) and a plasmon–phonon coupling extension (Photonics 2020). Neither has an open implementation. qwkit's Ando module is the entry point to both, and to the "beyond Ando" second-order paper.

Emerging platforms: oxides, nitrides and vdW wells have little self-consistent theory

Oxides: the best local fit

Oxide ISB optics has one firm experiment: room-temperature ISB absorption approaching 1.55 µm in SrTiO3/LaAlO3 MQWs, across a ~2.4 eV conduction-band offset (Ortmann et al., ACS Nano 2018).

BaSnO3 wells exist only as claims at APS-abstract, thesis and conference level: - "giant" $\chi(2)$ (APS 2022); - near-IR-to-visible ISB with $m^* \approx 0.2\ m_e$ (APS 2023); - field-tuned ISB (NUSOD 2022).

The underlying thesis measured no ISB absorption and lists nonparabolicity, phonons and χ(2) as untreated (Jang 2023).

The polar LaInO3/BaSnO3 interface has been studied only for transport. It shows: - a ~1.6 eV offset; - **~2100 cm²/V·s at 10 K**; - gate-tuned densities up to ~10^{14} cm^{-2} (the paper's figures are internally inconsistent on the exact value) (Lee et al. 2024); - a DFT-predicted 2DEG of up to 0.5 e per unit-cell area in the dispersive Sn-5s band (Aggoune & Draxl).

That makes LaInO3/BaSnO3 a polarisation-asymmetric optical quantum well with nonzero χ(2) and no compositional stepping, which is exactly qwkit's domain. Paik brings adsorption-controlled BaSnO3 MBE from the Schlom/PARADIM lineage to OU ECE (OU). A pure-theory precedent exists: β-(AlGa)2O3/Ga2O3 ISB and QWIP design predicted ~10^{11} Jones at 1.55 μm (Lyman & Krishnamoorthy 2020).

Nitrides and group IV

- **ScAlN/GaN.** Lattice-matched Sc0.14Al0.86N/GaN wells show ISB at 515–709 meV, and net polarisation rises with Sc "faster than previous theoretical predictions". This is a clean inverse problem, but Purdue is fitting its own data (Abdin et al. 2025).
- **GaN/AlN.** Telecom ISB is now combined with plasmonic antennas for SHG (Mundry et al.).
- **Deep-UV AlGaN.** The problems are p-doping and valence-band ordering, which need k·p (Jena 2026).
- **Ge/SiGe.** n-type L-valley Ge/SiGe gave the first group-IV THz ISB electroluminescence at 3.4 and 4.9 THz, with "suboptimal injection" (Stark et al. 2021). Asymmetric coupled Ge/SiGe wells have been engineered (Persichetti et al. 2020). Ge is nonpolar, so lifetimes need deformation-potential scattering, not Fröhlich.

Other platforms

- **ZnO.** m-plane ZnO/MgZnO emitted at **8.5 THz** up to room temperature, inside the GaAs reststrahlen gap, but no laser yet (Meng et al. 2021).
- **CdSe nanoplatelets.** They show 900–1600 nm ISB with atomically defined thickness (Diroll et al. 2019).
- **Few-layer van der Waals wells.** ISB has been tuned electrically by more than 100 meV in WSe2 (Wutz et al., ACS Nano 2026) and by more than 250 meV in InSe (Shcherbakov et al. 2024). Both use phenomenological infinite-well fits. No resonant χ(2) theory under displacement field was found.

Master table: every paper opportunity, de-duplicated and ranked

Legend

The columns use these labels: - **Reuse:** Full, Ext (extension) or New (see the qwkit section). - **Experiment:** No; Opt (optional proposed experiment written into the paper); Collab (needs a partner's data or devices). - **Prior-art status:** NF (not found in the notes' searches); Partial (named partial prior art); Incumbent (strong competing group); Unchecked (no search done). - **Effort, for one person:** S ≤3 months, M 3–6, L 6–12, XL >12. - **Priority:** P1 start within 12 months; P2 strong but second wave or collaborator-dependent; P3 opportunistic; P4 park or avoid.

Entries tagged [E] come from the earlier frontier survey. Merged sources appear in brackets.

ID	Title	Area	Gap and why it is open	Calculation	qwkit reuse	Needs experiment?	Prior-art check status	Target venue	Effort	Priority
NL-01	Interface and monolayer tolerance of bias-sign-reversal χ(2) designs	Nonlinear	Bank/Khurgin found interface abruptness limits engineered χ(2); no tolerance study for sign-reversal stacks	Monte Carlo over interdiffusion profiles and ±1 ML errors; χ(2) magnitude, sign, QPM duty error	Full	No	NF	Fold into N1, or Opt. Express / JOSA B	S	P1 (fold into N1)

ID	Title	Area	Gap and why it is open	Calculation	qwkit reuse	Needs experiment?	Prior-art check status	Target venue	Effort	Priority
NL-02	Saturation-limited efficiency and contrast bounds for ISB conversion in waveguides and polaritonic metasurfaces [E1 + switch/limiter bounds + model-hierarchy benchmark]	Nonlinear	Field uses two-level Maxwell–Bloch bounds (ΔR, Ω_max ≈ 0.26 ω_q); saturation caps total power; no multilevel bound	Multilevel saturation + DM χ(2) + TM modes + coupled-mode; compare two-level MB, multilevel DM, TDSP	Full (TDSP arm Ext)	No	NF (only two-level formulas in 2025 review)	PR Applied / Optica	M	P1
NL-03	Electrically programmable QPM for THz DFG, benchmarked against CW metasurface DFG [E2 + metasurface DFG theory]	Nonlinear	CW THz DFG from metasurfaces is conference-level only; no programmable-QPM waveguide THz design found	Three-level DM χ(2) at THz, THz free-carrier and phonon loss, segmented QPM	Ext (THz loss and dispersion)	Opt (Belkin/TUM-type)	Partial (QCL intracavity DFG lineage, unchecked)	PR Applied / APL Photonics	M	P2
NL-04	Mid-IR photon pairs from bias-programmed QPM [E6]	Nonlinear	Quantum ISB light is almost all theory; no ISB SPDC design found	SPDC joint spectral amplitude from χ(2) + QPM + waveguide dispersion; loss and thermal-noise floor	Ext (JSA, noise)	No	Unchecked (interband telecom pairs proposed 2026)	Optica Quantum / PR Applied	M	P2
NL-05	Collective (TDHF/RPA) second-order response beyond Ando: depolarisation detuning of double resonance [E4 + many-body χ(2)]	Nonlinear / Methods	Ando is a linear correction; TDSP shows intensity-dependent collective shifts; χ(2) in the multisubband-plasmon regime unverified	Second-order TDHF/RPA in subband basis; compare with DM χ(2) + Ando; TDSP spot checks	Ext	No	Partial (older Rosencher/Bois work unverified)	PRB	M	P1
NL-06	Floquet Stark-modulated ISB metasurfaces for nonreciprocity and frequency shifting	Nonlinear	Nonreciprocal ISB routers proposed but not demonstrated; RC limit 3.38 MHz	Time-modulated DM + coupled-mode; RC and capacitance from electrode solver	Ext	No	NF (only JoFET analogue)	Nanophotonics / ACS Photonics	M–L	P2
NL-07	Non-perturbative ISB harmonic generation (THG, 5HG, HHG) via multi-subband TDSP	Nonlinear	No QW-specific HHG found; Cominotti TDSP limited to the coherent shift	Real-time Schrödinger–Poisson with continuum and depolarisation	New (real-time solver)	No	NF (search-level)	PRB	L	P3
NL-08	Bandwidth and figure-of-merit bounds for ISB Stark modulators	Nonlinear	10 Gbit/s devices exist; no general FoM or bound paper	Stark coefficient vs insertion loss, tunnelling time, patch capacitance	Full	No	NF	PR Applied / IEEE JQE	S–M	P2
NL-09	Doubly resonant two-photon QWIP/QCD for ps autocorrelation of ring-QCL solitons	Nonlinear	68 GHz soliton crystals uncharacterised; 1 ps pulses need quadratic detectors	Two-photon rates, LO lifetimes, photoconductive gain	Ext (gain and transport)	Opt	Partial (two-photon QWIPs exist at HZDR)	PR Applied / Opt. Express	M	P2
NL-10	Microscopic Kerr, Bloch-gain lineshape and LEF design maps for ring-QCL solitons	Nonlinear	Bloch lineshape “not well understood”; DM can get the LEF sign wrong	k-resolved DM with counter-rotating terms, validated against published NEGF	Ext (transport DM)	No	Incumbent (TU Wien, ETH, Harvard)	PRL / PRB	L	P3
NL-11	Polariton–LO-phonon stimulated scattering design maps toward ISB polariton lasing	Nonlinear	Rate “extremely low”; no broad design-space study	Hopfield coefficients × Fröhlich rates over well and cavity parameters	Full / light Ext	No	Partial (Colombelli/Carusotto custom model)	PR Applied	M	P2

ID	Title	Area	Gap and why it is open	Calculation	qwkit reuse	Needs experiment?	Prior-art check status	Target venue	Effort	Priority
NL-12	Materials-specific Rydberg-like ISB polariton blockade (Oliveira–Ciuti mechanism)	Nonlinear	Proposal is model-level; no U/γ estimate for real wells	Bound-to-continuum oscillator strengths, Bohr radius, U/γ for InGaAs, GaAs and InAs/AlSb in patch cavities	Ext (discretised continuum)	No	NF; originators may follow up	PRB / PRL	M	P2 (time-sensitive)
NL-13	Squeezed light from QCLs with sub-shot-noise pumping: cascade Langevin model	Nonlinear	Excess noise and detector mismatch block quantum QCL light	Multilevel rate equations + Langevin noise; compare published data	Ext	No	Partial (Grillot ICL models)	PR Applied / Opt. Express	M	P3
NL-14	Near-IR transfer of sign-reversal QPM: InAs/AlSb and GaN/AlN coupled wells, including low-Isat designs	Nonlinear / Materials	GaN Isat 9.6 vs 3.9 W/μm² theory; claimed 30× reduction undemonstrated; InAs/AlSb triple well unexplored	Reparameterisation; nitride polarisation fields; Isat and QPM efficiency	Ext (parameters, sheet charge)	No	Partial (older nitride simulations)	Opt. Express / JAP	M	P2
NL-15	Bias-switchable interband χ(2) sign in mirror-symmetric coupled wells for electrically poled 1550→775 nm QPM	Nonlinear	Bank/Khurgin designs are fixed-asymmetric (2750 pm/V); no switchable-sign interband scheme found	Multiband k·p interband χ(2) with excitons	New (k·p)	No (Collab upside with Bank)	NF; Incumbent risk high	Optica / APL Photonics	L	P2
NL-16	Transmissive and all-dielectric ISB nonlinear metasurfaces	Nonlinear	Review asks for transmissive, low-loss designs	RCWA/FDTD + DM χ(2)	New	No	Partial	Nanophotonics	L	P4
MS-01	qwkit: an open, validated intersubband-optics chain in Python (with tutorial notebooks)	Methods	No open code combines depolarisation, LO rates, χ(2), saturation and waveguides	Docs, tests, validation against published spectra and lifetimes	Full	No	NF for this feature set	JOSS (short), CPC (long)	S	P1
MS-02	Differentiable, fluctuation-aware inverse design of bias-switchable heterostructures [E3 + autodiff SP + robust design + χ(2) surrogate]	Methods / Nonlinear	No autodiff SP found; ML surrogates are local and fluctuation-blind	JAX port; adjoint eigen-derivatives through nonparabolicity and self-consistency; robust objective; compare with BO and MLP	Ext (JAX port)	No	NF (BO 2020, MLP 2024, ANN 2026 bound novelty)	npj Comput. Mater. / PR Applied	M–L	P1
MS-03	Global sensitivity and UQ of QW observables to band and interface-roughness parameters	Methods	No UQ of standard parameter spreads found	Sobol / polynomial chaos on ISB energy, dipole, χ(2), LO lifetime, roughness linewidth	Full (roughness linewidth Ext)	No	NF	JAP / Semicond. Sci. Technol.	S–M	P1 (or fold into MS-02)
MS-04	Cross-code benchmark and open labelled heterostructure dataset [benchmark + dataset + transport suite]	Methods	No versioned benchmark or ML dataset	Same structures across qwkit, Aestimo, kdotpy, ErwinJr2; Zenodo release	Full	No	NF	J. Comput. Electron. / Sci. Data	M	P2
MS-05	Open multisubband-plasmon solver with plasmon–LO-phonon coupling and nonparabolicity	Methods	Theory by Paris group; no open code	Dipole-representation Hamiltonian, Fröhlich coupling, multiple wells	Ext	No	Partial (Photonics 2020)	PRB / JOSS	M	P2

ID	Title	Area	Gap and why it is open	Calculation	qwkit reuse	Needs experiment?	Prior-art check status	Target venue	Effort	Priority
MS-06	Unified non-Lorentzian ISB lineshapes (many-body, inhomogeneous, nonparabolic)	Methods	Old fragments, no unified open model	Lineshape model fitted to published spectra	Ext	No	Partial	JAP / PRB	M	P3
MS-07	Reduced-order e–e scattering module for ISB density-matrix models	Methods	Most codes stop at Hartree	Hartree vs simplified NEGF-like vs Boltzmann e–e	Ext	No	Partial	J. Comput. Electron.	M	P3
MS-08	Spurious-solution-free 8/14-band k·p benchmark and decision guide	Methods	Handled ad hoc per paper	Multiple discretisations on standard structures	New (k·p)	No	Partial (piecemeal)	J. Comput. Electron. / OQE	L	P3 (prerequisite for NL-15)
MS-09	Continuum-plus-disorder vs atomistic TB for InGaN QWs	Methods	Accuracy map vs TB missing	Landscape theory in 2D/3D	New	No	Partial	PRB	L	P4
MS-10	Open QCL NEGF code	Methods	No open QCL NEGF	Full NEGF	New	No	Incumbent (commercial and group codes)	—	XL	P4 (avoid)
SC-01	Escape-to-device model for 80–300-period strain-balanced superlattices [device model + escape-rate theory]	Solar	~10 mV unexplained Voc loss; 5 nm tunnel/thermionic crossover; drift-diffusion judged inadequate	SP + thermionic, tunnelling and LO-assisted escape + capture + SRH/radiative + photon recycling + drift-diffusion	Ext + New (holes, interband, drift-diffusion)	No (Collab upside: France, Sellers)	Partial (Connolly, Toprasertpong, Aeberhard NEGF)	JAP / IEEE JPV / Prog. Photovolt.	L	P2
SC-02	Hot-carrier thermalisation in type-II InAs/AlAsSb QWs with real band structure and hot-phonon kinetics [hot-carrier solar + phonon bottleneck]	Solar	Existing kinetic model is generic type-I; pathways debated	Nonparabolic type-II SP, LO/interface-phonon rates, hot-phonon non-equilibrium, overlap-limited radiative rate → Tc(P), Q	Ext (holes, type-II, phonon reabsorption)	No (Collab upside: Santos, Mishima)	NF for type-II; check Ferry/ASU, Cavassilas	PR Applied / JAP / J. Phys. D	M–L	P1
SC-03	Hot-carrier MQW limit with subband-designed energy-selective contacts	Solar	SQW contact gave Voc 0.28 V without hot-carrier operation; level needs redesign	Resonant-tunnelling transmission + detailed balance using SC-02 thermalisation coefficient	Ext (transfer matrix)	No	Partial (Conibeer, Guillemoles)	EPJ PV / Prog. Photovolt.	M	P2
SC-04	Reciprocity and ERE breakdown in MQW solar cells	Solar	Aeberhard treated bulk ultrathin cells; record QW papers report no ERE	EQE(V) and EL(V) with QCSE, η_coll(V), TE/TM emission; Rau-error map	Ext (interband, escape)	No	NF for MQW	PR Applied	M–L	P2
SC-05	Field-dependent two-step photon upconversion: escape vs tunnelling and the TM intraband step	Solar	Kobe lists optimal field and extraction balance as open; 5.1% vs 46.8%	SP + tunnelling escape + DM ISB (TM) + detailed balance; QW vs QD	Ext	No	NF	JAP / PR Applied	M	P2
SC-06	QW intermediate-band and ratchet limits with realistic ISB absorptivity (depolarisation, filling)	Solar	Limits use idealised step absorptivities despite strong filling dependence	DM ISB + Ando + self-consistent filling inside detailed balance	Ext (light)	No	Unchecked vs Imperial theses, Luque/Martí	PR Applied / Prog. Photovolt.	M	P2 (highest reuse in solar)
SC-07	End-of-life radiation modelling of QW space cells	Solar	QW cells keep Jsc but retain 60% vs 72% efficiency	Poisson/drift-diffusion with fluence-dependent carrier removal and lifetime; QW placement	New (drift-diffusion)	No	NF	IEEE JPV / SEMSC	M	P3

ID	Title	Area	Gap and why it is open	Calculation	qwkit reuse	Needs experiment?	Prior-art check status	Target venue	Effort	Priority
SC-08	Inverse design of strain-balanced QW stacks for current matching (AM1.5G, AM0, EOL; III–V/Si middle cell)	Solar	No QW-stack inverse design found	SC-01 model + MS-02 optimiser	New (after SC-01)	No	NF	Solar RRL / IEEE JPV	M	P3
SC-09	Luminescent coupling in GaInP/GaAs-QW tandems	Solar	QW edge 1.351 eV below host; no QW-specific LC study	Coupling fractions and mismatch tolerance	New (simple)	No	NF	IEEE JPV	S–M	P3
SC-10	Review and open dataset: strain-balanced QW cells after 39.5%	Solar	No post-2022 review	Meta-analysis of Voc, Woc, per-well Jsc	n/a	No	NF	Prog. Photovolt. / JAP Tutorial	M	P3
SC-11	Valley-photovoltaic heterointerface theory	Solar	OU's valley PV was limited by degeneracy mismatch	Multivalley transport and interface design	New	Opt (OU MBE)	Unchecked	PR Applied	L	P3
SC-12	2D Ruddlesden–Popper perovskites as MQW absorbers: n-dependent PV limit	Solar	SCAPS-only device papers; mature exciton theory elsewhere	Image-charge EMA + detailed balance	New	No	Unchecked (large exciton literature)	Solar RRL	M	P4
TPV-01	QW absorbers in air-bridge, high-reflectance TPV: detailed balance with Auger and photon recycling	TPV	Thin absorbers win under recycling; no 2018–2026 QW re-analysis	Interband QW absorption + Auger + reflector cavity, vs bulk 0.74/0.9 eV	Ext + New (interband, Auger)	No	NF	PR Applied / JAP	M–L	P2
TPV-02	Quantum-corrected coupled fluctuational electrodynamics + transport for QW near-field cells	TPV	Coupled FE + drift-diffusion exists for bulk only	FE + SP + capture/escape	New (FE)	No	NF	PR Applied	L	P3
TPV-03	Interband-cascade TPV: near-field, thermoradiative and negative-luminescence operation	TPV	Near-field IC-TPV and IC thermoradiative modelling not found	Per-stage current matching, recycling, Auger; absorption from Yang group or 8-band k·p	New (or borrowed absorption)	Collab (R. Q. Yang)	Partial (SPIE limit papers)	PR Applied / IEEE JPV	L	P2
TPV-04	QW vs double-heterostructure active regions for thermophotonic LEDs and electroluminescent cooling	TPV	Requirements paper omits QWs; EQE > 90–95% needed	Radiative, Auger and SRH balance, extraction, near-field QW dipole coupling	Ext + New	No	Partial (check Aalto parametric LED study)	PR Applied / SEMSC	M–L	P2
TPV-05	Temperature limits of narrowband, bias-switchable ISB thermal emitters for low-temperature TPV and TPX [TPV emitter + E7]	TPV	Noda switched 0.74→0.24 near room temperature; no 500–1000 K theory	DM + Ando + thermal subband population + LO dephasing + nonparabolicity + TM cavity; Kirchhoff emissivity into detailed balance with IC-TPV/T2SL cells	Full (+ thermal populations)	No (Opt: Yang IC cells)	NF	ACS Photonics / PR Applied	M	P1
TPV-06	Hot-carrier TPV with QW LO-phonon cooling and hot-phonon bottleneck	TPV	Nimje et al. use idealised cooling	Feed SC-02 rates into hot-carrier TPV detailed balance	Ext (after SC-02)	No	NF	PR Applied	S–M	P2

ID	Title	Area	Gap and why it is open	Calculation	qwkit reuse	Needs experiment?	Prior-art check status	Target venue	Effort	Priority
TPV-07	What limits out-of-band reflectance above 99%: free-carrier and intervalence absorption in doped layers	TPV	Ge model names FCA dominant; no general design rules	Multilayer TMM + Drude/IVBA	Ext (TMM)	No	Partial (Ge 2026 model)	SEMSC / JAP	S–M	P3
TPV-08	Zero-gap high-index-spacer TPV with QW or low-gap cells	TPV	Concept and a-Si modelling done	Conduction vs radiation trade-off	New	No	Partial	IJHMT / JQSRT	M	P4
TPV-09	Tutorial: quantum structures in thermal-photonic energy conversion	TPV	No QW-focused review	Synthesis	n/a	No	NF	JAP Tutorial / J. Photonics Energy	M	P3
MAT-01	ISB absorption, χ(2) and LO-phonon lifetimes at polar LaInO3/BaSnO3 interfaces and superlattices vs EDL-gated density	Materials	Interface studied only for transport; BaSnO3 ISB/χ(2) only at abstract or thesis level	SP with polarisation sheet charge, nonparabolic Sn-5s band, Ando, DM χ(2), Fröhlich, parameter-uncertainty bands	Ext (parameters, sheet charge, density-dependent mass)	Opt (Paik MBE; Venkatesan PLD)	NF; check Demkov and Belgrade follow-ups	PR Applied / PR Materials / APL Materials	M	P1
MAT-02	SrTiO3/LaAlO3 MQW ISB with multi-orbital masses and field- and temperature-dependent permittivity	Materials	Measured 2018; no temperature or ε(E)-resolved prediction	Per-orbital SP; nonlinear Poisson	Ext (nonlinear Poisson)	Opt (Venkatesan)	Partial (Ortmann 2018 theory)	PRB / PR Materials	M	P2
MAT-03	Self-consistent ISB and resonant χ(2) in displacement-field-gated few-layer InSe and WSe2 [gated vdW χ(2)]	Materials / Nonlinear	Experiments use infinite-well fits; no χ(2) theory found	Layer-chain/EMA SP, per-valley m_z, Ando, DM χ(2), waveguide modulator	Ext (layer chain)	No	NF (search-level)	PRB / 2D Mater.	M	P2
MAT-04	β-(AlGa)2O3/Ga2O3 wells: multi-mode polar-phonon lifetimes, lineshape, χ(2), QCL feasibility	Materials	Energies and QWIP metrics done (2020); lifetimes and χ(2) not	Multi-mode anisotropic Fröhlich sum	Ext	No	Partial	JAP / APL Materials	M	P2
MAT-05	ScAlN/GaN polarisation refit from published ISB data, plus polarisation-robust QCL and χ(2) designs	Materials	Measured polarisation rises faster than theory	Global P_sp(x) fit with UQ; robust designs	Ext (nitride parameters)	No	Incumbent (Purdue fitting own data)	APL / JAP	M	P3
MAT-06	m-plane ZnO/MgZnO THz QCL design at 8–9 THz	Materials	EL demonstrated; no laser	DM transport design	Ext (QCL transport)	No	Incumbent (Faist group)	PR Applied / JAP	L	P3
MAT-07	CdSe nanoplatelet ISB vs monolayer count: nonparabolicity and dielectric benchmark	Materials	Measured 2019; rigorous EMA fit unverified	Kane EMA + ligand barrier + image potential + Ando	Ext (light)	No	Unchecked	J. Phys. Chem. C / PRB	S	P2
MAT-08	Ge/SiGe symmetric coupled wells: bias-switched χ(2) for CMOS-compatible THz/mid-IR [E5]	Materials / Nonlinear	Ge has no bulk χ(2); EL shown 2021; no switchable-χ(2) design checked	L-valley SP + deformation-potential and intervalley scattering	New (scattering)	No	Unchecked	PR Applied	M–L	P2

ID	Title	Area	Gap and why it is open	Calculation	qwkit reuse	Needs experiment?	Prior-art check status	Target venue	Effort	Priority
MAT-09	GeSn/SiGeSn ISB parameter-sensitivity map	Materials	Γ/L ordering and offsets uncertain	UQ across literature parameters	Ext	No	Unchecked	JAP	M	P3
MAT-10	SrVO3/SrTiO3 metallic-QW ENZ and plasmon tuning	Materials	ARPES QW states known; optics not searched	Drude/hydrodynamic with renormalised mass	New	Opt (Paik/Venkatesan)	Unchecked	ACS Photonics	M	P4
MAT-11	KTaO3, EuO, HgTe, deep-UV AlGaN valence physics, InAs/GaSb T2SL interband absorption	Materials	Need k·p or DFT beyond a 1D single-band code	—	New	—	—	—	XL	P4 (avoid without k·p)

Count, merges and collaboration hooks

The table holds 58 entries: 51 actionable papers plus 7 parked or avoided directions (NL-16, MS-09, MS-10, SC-12, TPV-08, MAT-10, MAT-11). Of the 51, six list an optional proposed or collaborative experiment as their primary mode (NL-03, NL-09, SC-11, TPV-03, MAT-01, MAT-02). The other 45 need no experiment, though several note a collaborative upside.

All seven earlier frontier-survey ideas survive as merged entries: - E1 → NL-02 - E2 → NL-03 - E3 → MS-02 - E4 → NL-05 - E5 → MAT-08 - E6 → NL-04 - E7 → TPV-05

Several papers pair theory with a proposed or collaborative experiment. Each is written as a theory paper with a measurement-ready prediction section: - **MAT-01** (Paik): unit-cell-exact BaSnO3/LaInO3 recipes and predicted Brewster-angle FTIR spectra vs gate density. - **MAT-02** and **MAT-10** (Venkatesan): PLD SrTiO3/LaAlO3 and SrVO3. - **TPV-03** and **TPV-05** (Rui Q. Yang): IC-TPV cells as receivers for ISB emitters. - **SC-02**, **SC-03** and **SC-11** (OU MBE, Santos and Mishima): hot-carrier InAs/AlAsSb and valley structures. - **SC-01** (France and Sellers): published and unpublished MQW data. - **NL-03** and **NL-09** (Belkin/TUM, Lee/UNIST-type metasurface groups). - **NL-15** (Bank).

A six-paper programme from toolkit to OU growth

Why these six

The sequence below takes the highest-ranked papers that can start with qwkit as it stands. Each builds a module or result the next one needs. N1 sits in front: clear its parked checks and submit it, ideally folding NL-01’s tolerance analysis in as a robustness section.

Order	ID	Paper	Why it sits here	qwkit reuse	Approx. months (solo, overlapping)	Venue	OU or external hook
0	N1 (+NL-01)	Bias-polarity sign-reversal QPM with mirror-symmetric triple wells	Complete; needs the prior-art checks below	Full	0–2	As planned	—
1	MS-01	qwkit software paper	Makes every later paper reproducible and citable; low effort, fast review	Full	0–3	JOSS	—
2	NL-02	Saturation-limited efficiency and contrast bounds, waveguide and metasurface	Direct sequel to N1’s saturation-aware comparison; reuses every module; answers the review’s two-level gap	Full	2–7	PR Applied / Optica	Belkin/Lee-type metasurface data for comparison

Order	ID	Paper	Why it sits here	qwkit reuse	Approx. months (solo, overlapping)	Venue	OU or external hook
3	NL-05	Second-order TDHF/RPA beyond Ando	Tests whether NL-02's high-doping bounds survive collective effects; physics depth for a PRB	Ext	6–11	PRB	—
4	MS-02 (+MS-03)	Differentiable, fluctuation-aware inverse design of bias-switchable stacks	Uses NL-02 and NL-05 objectives; JAX port speeds every later study; releases the dataset	Ext	9–15	npj Comput. Mater. / PR Applied	—
5	TPV-05	Temperature limits of switchable ISB thermal emitters in low-temperature TPV	First energy paper with near-full reuse; bridges to IC-TPV	Full	13–17	ACS Photonics / PR Applied	Rui Q. Yang (optional proposed test)
6	MAT-01	LaInO3/BaSnO3 polar-interface ISB and χ(2)	Moves the whole chain to a new material with local growth; ends in a proposed experiment	Ext	15–21	PR Applied / APL Materials	Paik (MBE), Venkatesan (PLD)

How the sequence fits together

Every step reuses the same validated chain, so the cumulative investment stays near one codebase. The programme reads as one argument: - **Tool:** an open tool (1). - **Limits:** what bias-programmable ISB nonlinear devices can ultimately do (2). - **Physics:** whether many-body effects move those limits (3). - **Design:** how to design stacks that reach them under real growth tolerances (4). - **Energy:** the same physics as a thermal emitter (5). - **Materials:** a new oxide platform grown at OU (6).

Swap-ins

SC-02 (hot-carrier InAs/AlAsSb) is the first alternate. It is P1 on novelty and local fit, but it needs hole and type-II extensions. Run it in parallel with step 4, or let it replace step 6 if Paik's group cannot commit to BaSnO3/LaInO3 growth within the window.

The second wave, from about month 18, should build the shared interband/escape module, ideally by coupling to Solcore's interband absorption. That module unlocks the solar and TPV device cluster: SC-04, SC-05, SC-06, TPV-01 and SC-01.

Time-sensitive items can jump the queue if the originating groups go quiet: NL-12 (Oliveira–Ciuti materials estimate) and NL-15 (interband switchable χ(2)).

Verification checks still required before each paper

N1 and the six sequence papers

Each row lists the checks that decide go or no-go. All "not found" claims need a Scopus or Web of Science search plus forward citations of the anchor papers named.

Paper	Prior-art checks	Inputs to gather	Go / no-go criterion
N1	Read the full text of Yu et al., Sci. Adv. 11, eadw8852 (2025): does it reverse the χ(2) sign by bias, and is it propagation QPM or local metasurface phase? Cite Park et al., LSA 13, 169 (2024) for electrically written phase gratings. Restate the US 4,880,297 / US 5,289,309 differentiation. Cite CLEO 2024 CW THz DFG if THz is discussed.	Interdiffusion and ±1 ML tolerance runs (NL-01) for a robustness section	Go if Sci. Adv. 2025 is local phase control, not waveguide QPM with a triple-well design; otherwise reframe novelty around saturation-aware efficiency
MS-01	JOSS scope and "substantial scholarly effort" criteria; overlap with Aestimo, ErwinJr2, kdotpy feature sets	Public repository, test suite, documentation, validation cases against published spectra and LO lifetimes	Go if the integrated depolarisation → χ(2) → waveguide chain is demonstrably absent from existing codes

Paper	Prior-art checks	Inputs to gather	Go / no-go criterion
NL-02	Forward citations of Krakofsky 2025 and Mann et al. Optica 2021; arXiv 2403.15911 full text; any 2025–2026 multilevel saturation-bound paper	Published metasurface saturation curves (ΔR vs intensity) and waveguide SHG data for validation	Go unless a multilevel, nonparabolic bound paper already exists
NL-05	Rosencher/Bois-era χ(2) many-body papers; forward citations of Cominotti PRB 2023 and the Montes Bajo JAP 2023 perspective	High-doping χ(2) or SHG spectra to show where Ando-level DM fails	Go if no second-order TDHF/RPA ISB treatment with a depolarisation-detuning result exists
MS-02	Forward citations of the BO-NEGF 2020, MLP 2024 and Stojković 2026 papers; search “differentiable Schrödinger–Poisson”, “JAX heterostructure”, “adjoint eigenvalue quantum well”	Benchmark targets from those papers to compare on identical objectives	Go unless an autodiff SP heterostructure optimiser is published
TPV-05	Noda-group 2015–2026 follow-ups; GaN/AlGaN ISB emitters; “intersubband” + “thermophotovoltaic”; thermal-emission reviews	Noda 2014 device structure for validation; IC-TPV and T2SL cell EQE (Yang group, Montpellier)	Go if no theory of high-temperature ISB emitter linewidth and switching contrast exists
MAT-01	Google Scholar: “Li Demkov BaSnO3 nonlinear” (2022–2026), “Stanojević BaSnO3 Optical and Quantum Electronics”, “LaInO3 BaSnO3 intersubband”; PARADIM 2024–2026 stannate superlattices	BaSnO3 and LaInO3 ε∞, ε0, LO energies; density-dependent BaSnO3 mass; LaInO3 bound charge; growth-realistic thicknesses from Paik	Go if no peer-reviewed BaSnO3-QW χ(2) paper with nonparabolicity and phonons exists and Paik confirms the structures are growable

Remaining P1 and P2 papers

The other P1/P2 papers need lighter but specific checks before work starts: - **SC-02:** search for NEGF or Boltzmann InAs/AlAsSb models by Ferry (ASU) and Cavassilas (Marseille), and for forward citations of Makhfudz 2022. Extract the measured thermalisation coefficients from the OU papers’ full text. - **SC-01:** read the Joule 2022 and Ahmed 2025 full texts for fitted escape and recombination rates. Search forward citations of France 2022. - **SC-04:** check forward citations of Aeberhard PRL 2017 for MQW extensions. Search for a Sugiyama-group QFLS/EL analysis after 2018. - **SC-05:** look for Kobe 2023–2026 TPU follow-ups beyond the JSAP 2024 abstracts. - **SC-06:** check the Imperial QW-IB and photon-ratchet theses and the Luque/Martí reviews. - **SC-03:** read the full Boyer-Richard 2024 design parameters. - **TPV-01:** search “quantum well” + “air-bridge” TPV. - **TPV-03:** confirm Rui Q. Yang’s 2022–2026 IC-TPV publications and whether the group will share absorption spectra. - **TPV-04:** read the Aalto “parametric study of highly-efficient LEDs for thermophotonic cooling”. - **TPV-06:** search forward citations of Nimje 2023. - **NL-03:** check QCL intracavity DFG theory and the CLEO 2024 journal follow-up. - **NL-04:** search “intersubband spontaneous parametric down-conversion”. - **NL-06:** check time-modulated ISB metasurfaces after Taravati 2024. - **NL-08:** check the Sirtori-group modulator papers’ bandwidth analyses. - **NL-09:** read the HZDR two-photon QWIP papers. - **NL-11:** check Colombelli/Carusotto design work after Goulain 2025. - **NL-12:** check Ciuti-group follow-ups to arXiv 2510.24421. - **NL-14:** look for InAs/AlSb or GaN/AlN coupled-well QPM proposals. - **NL-15:** check the latest Bank/Khurgin and Capasso 2026 outputs. - **MS-03:** search “uncertainty quantification” + “k·p” + “intersubband” in Scopus. - **MS-04:** check JOSS/Zenodo for existing heterostructure benchmarks. - **MS-05:** confirm no open multisubband-plasmon code exists from the Paris group. - **MAT-02:** read the Ortmann 2018 full text for its theory model. - **MAT-03:** search “second harmonic intersubband InSe few-layer theory”. - **MAT-04:** search Ga2O3 QW LO-phonon lifetime and χ(2) papers from 2021–2026. - **MAT-07:** check Diroll 2019 follow-ups for nonparabolic EMA fits. - **MAT-08:** search Ge/SiGe ISB χ(2)/DFG prior art.

P3 and P4 entries need the same database search, plus a competitor scan, only if promoted.

Conclusion

The mapping changes one strategic premise. The researcher’s comparative advantage is intersubband physics carried from microscopic rates all the way to propagation, not quantum wells in general. The fields with the biggest headline numbers, 39.5% solar and 44% TPV, are where qwkit currently reaches least. The places where theory clearly lags experiment are polaritonic metasurfaces, switchable thermal emitters and oxide or vdW wells with only phenomenological fits. These are where qwkit already reaches furthest, and where a solo theorist can publish fastest with the least competition.

The six-paper sequence turns that advantage into a body of work that ends in OU growth proposals. Building one shared interband module in year two then turns the energy threads from expensive one-offs into a cluster of four or five papers. They would share code, validation data and OU experimental partners: Santos/Mishima, Yang and Paik. The main risk is not running out of ideas but being scooped on a few time-sensitive ones. Four groups are working close to these threads: Belkin/Lee on electrical $\chi(2)$ control, Ciuti on ISB blockade, Bank/Khurgin on interband $\chi(2)$, and Purdue on ScAlN. So the database checks above should run before each start date, not after.